

Seneca

SEG800

Digital Signal and Image Processing

Other Transforms

Seneca College

Overview

- Other important transforms
 - DFT
 - DCT
 - FFT

Figure 7.1.1 Frequency-domain sampling of the Fourier transform.

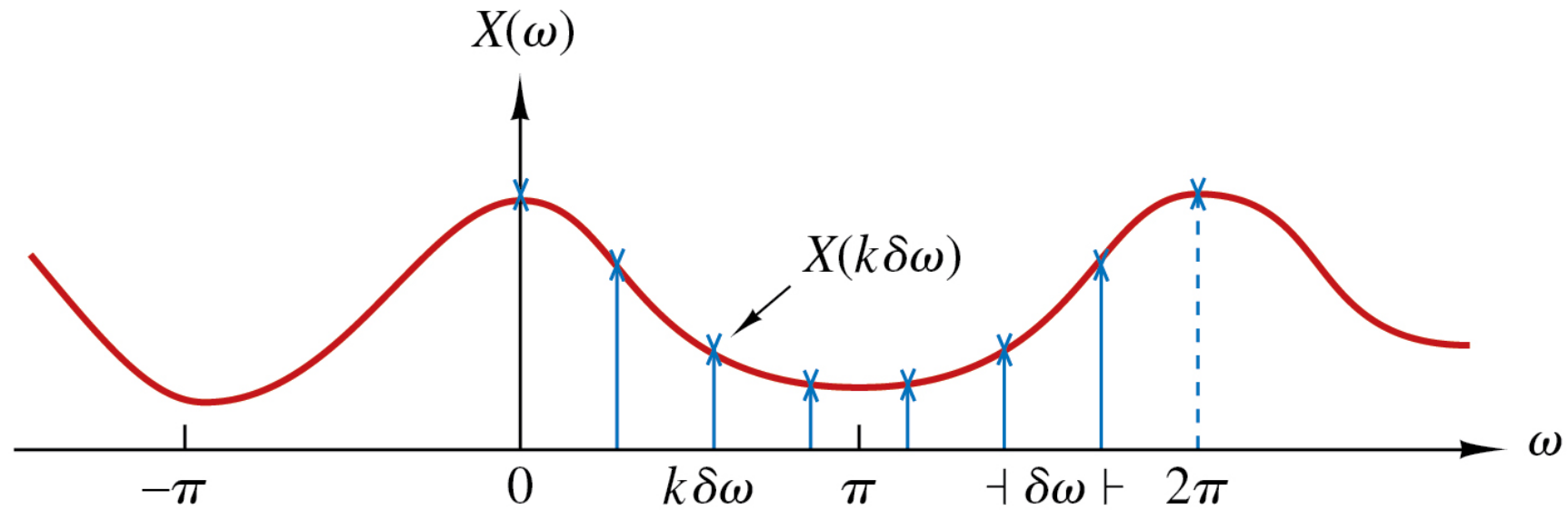


Figure 7.1.2 Aperiodic sequence $x(n)$ of length L and its periodic extension for $N \geq L$ (no aliasing) and $N < L$ (aliasing).

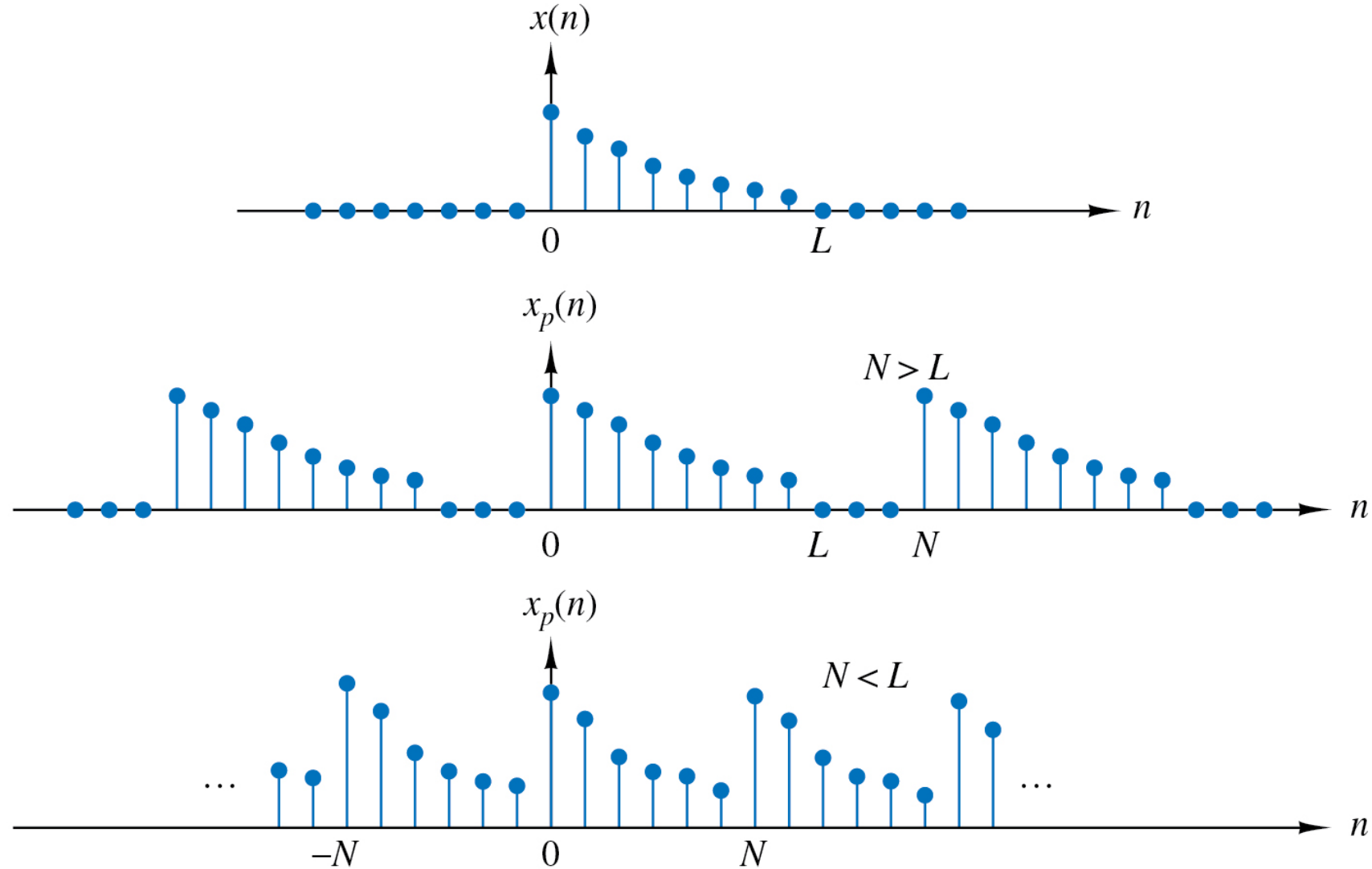


Figure 7.1.3 Plot of the function $[\sin(\omega N/2)]/[N \sin(\omega/2)]$.

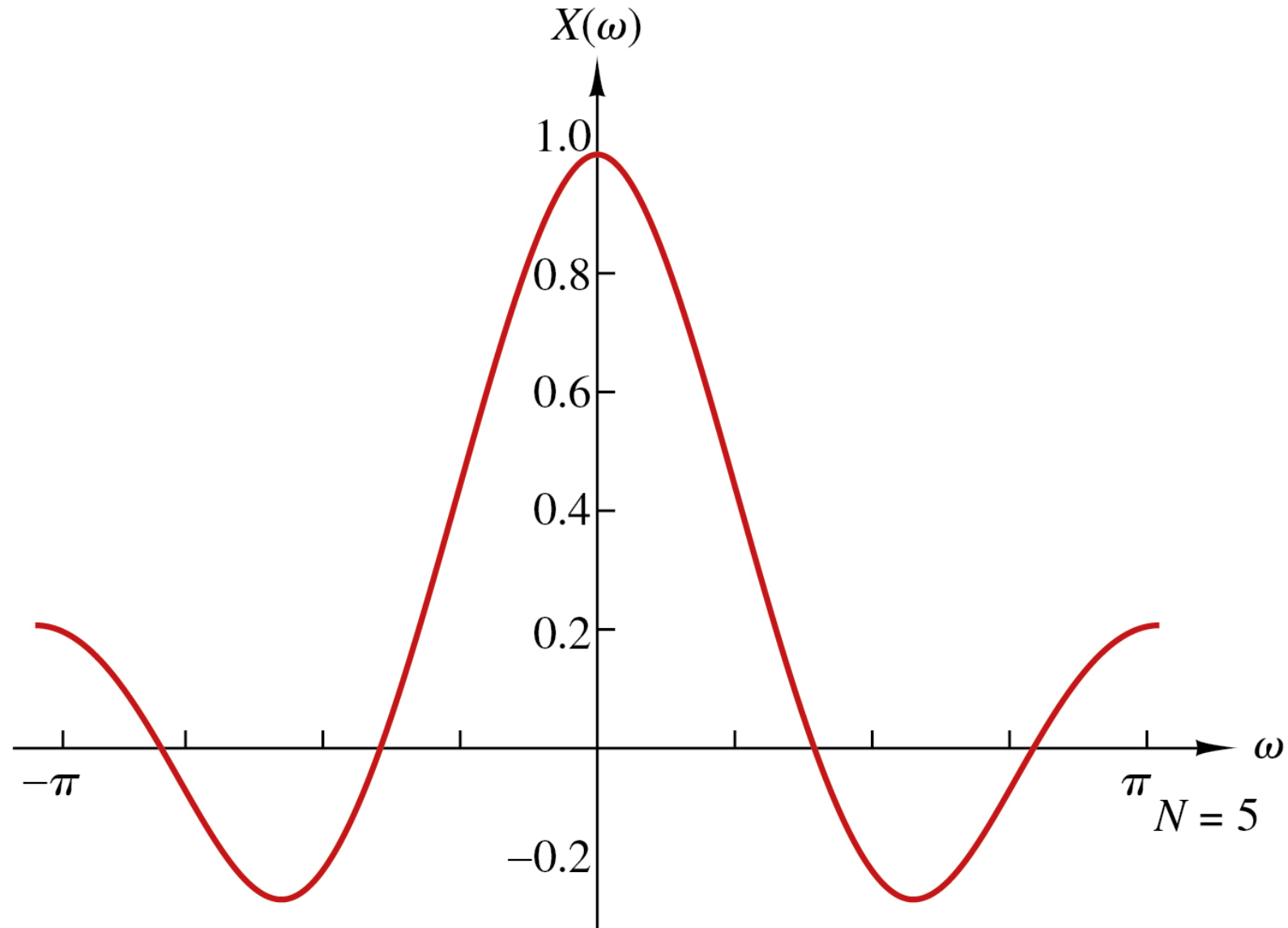


Figure 7.1.4 (a) Plot of sequence $x(n) = (0.8)^n u(n)$; (b) its Fourier transform (magnitude only); (c) effect of aliasing with $N = 5$; (d) reduced effect of aliasing with $N = 50$.

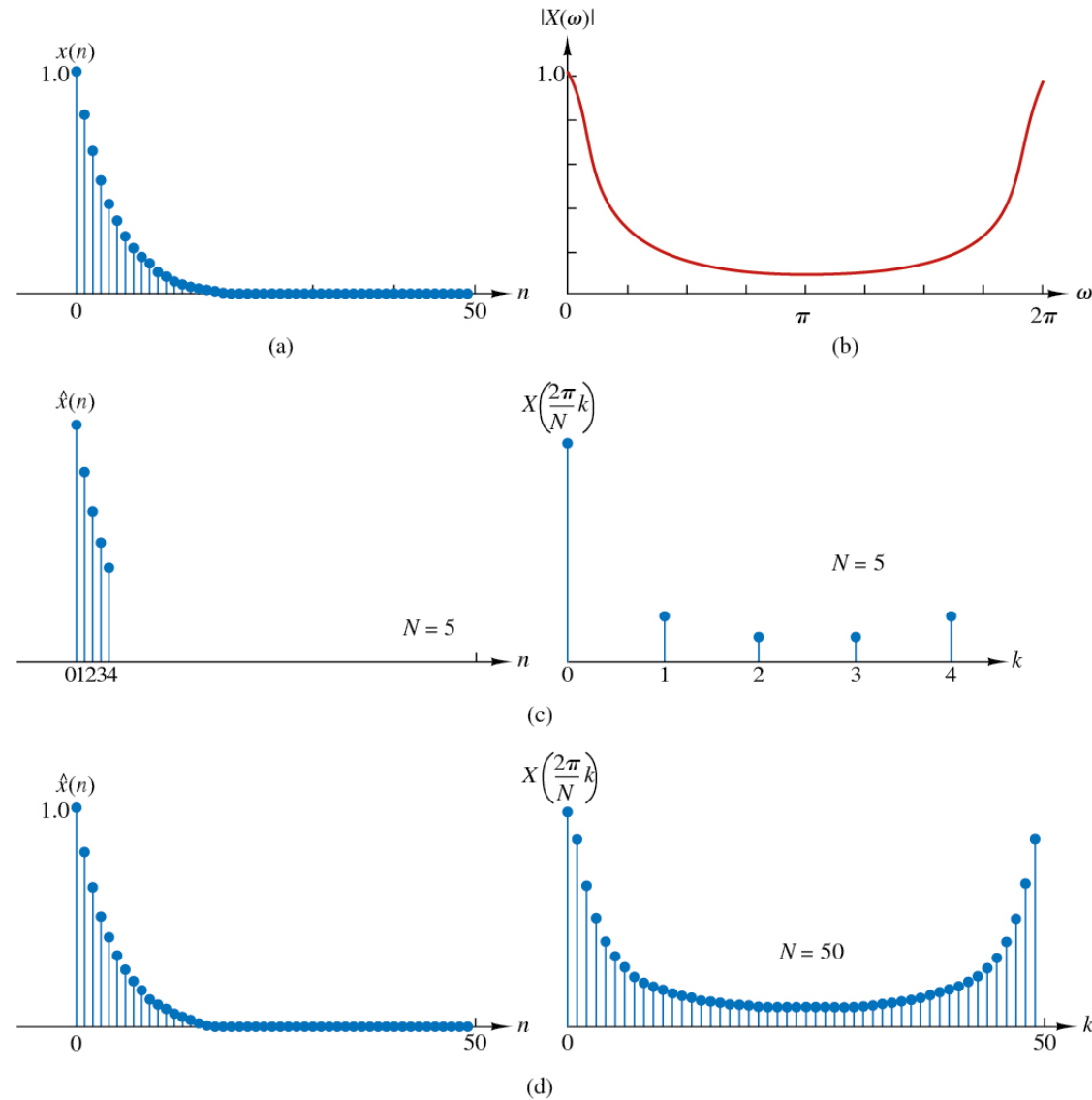


Figure 7.1.5 Magnitude and phase characteristics of the Fourier transform for the signal in Example 7.1.2.

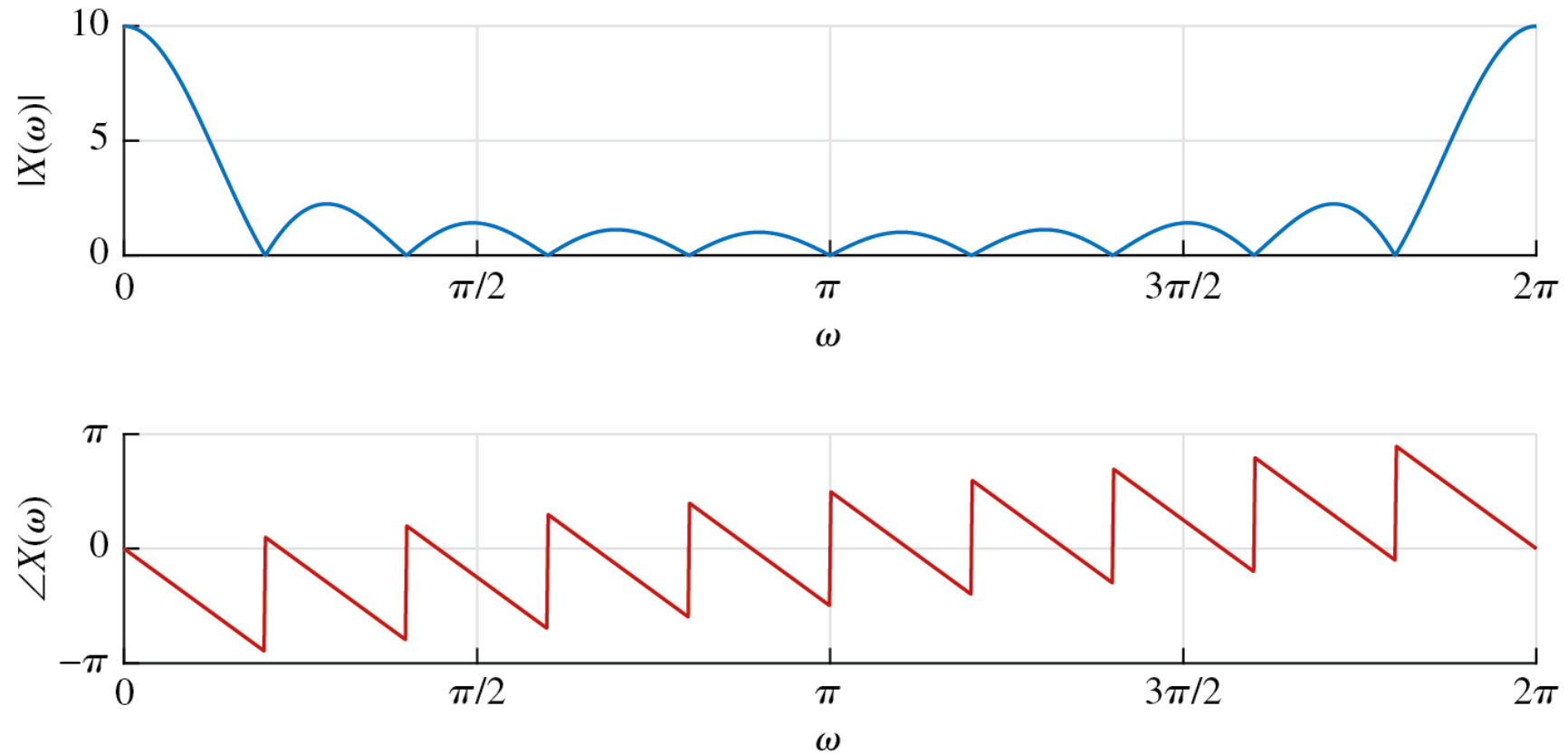


Figure 7.1.6 Magnitude and phase of an N -point DFT in Example 7.1.2;
 (a) $L = 10, N = 50$; (b) $L = 10, N = 100$.

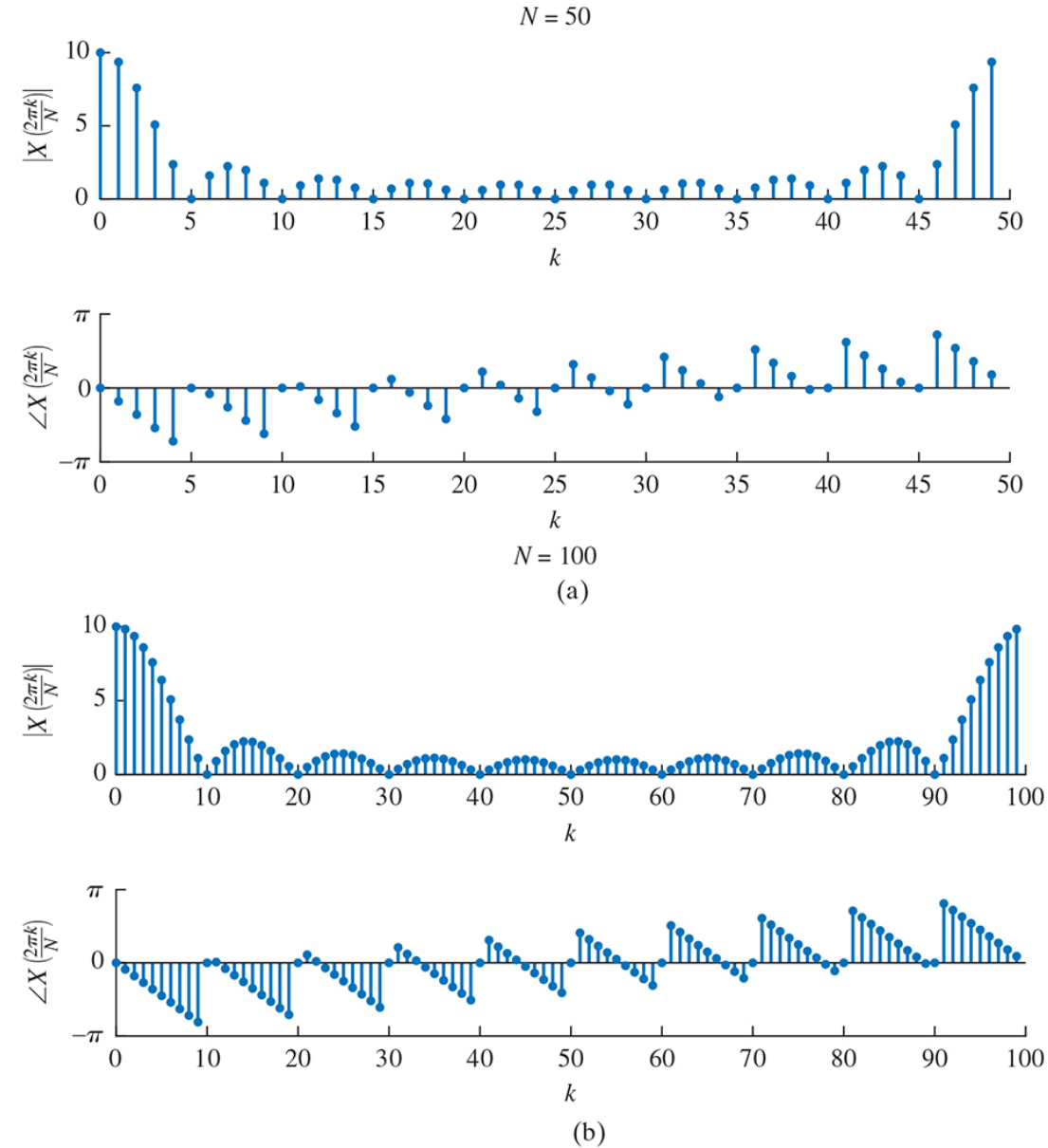
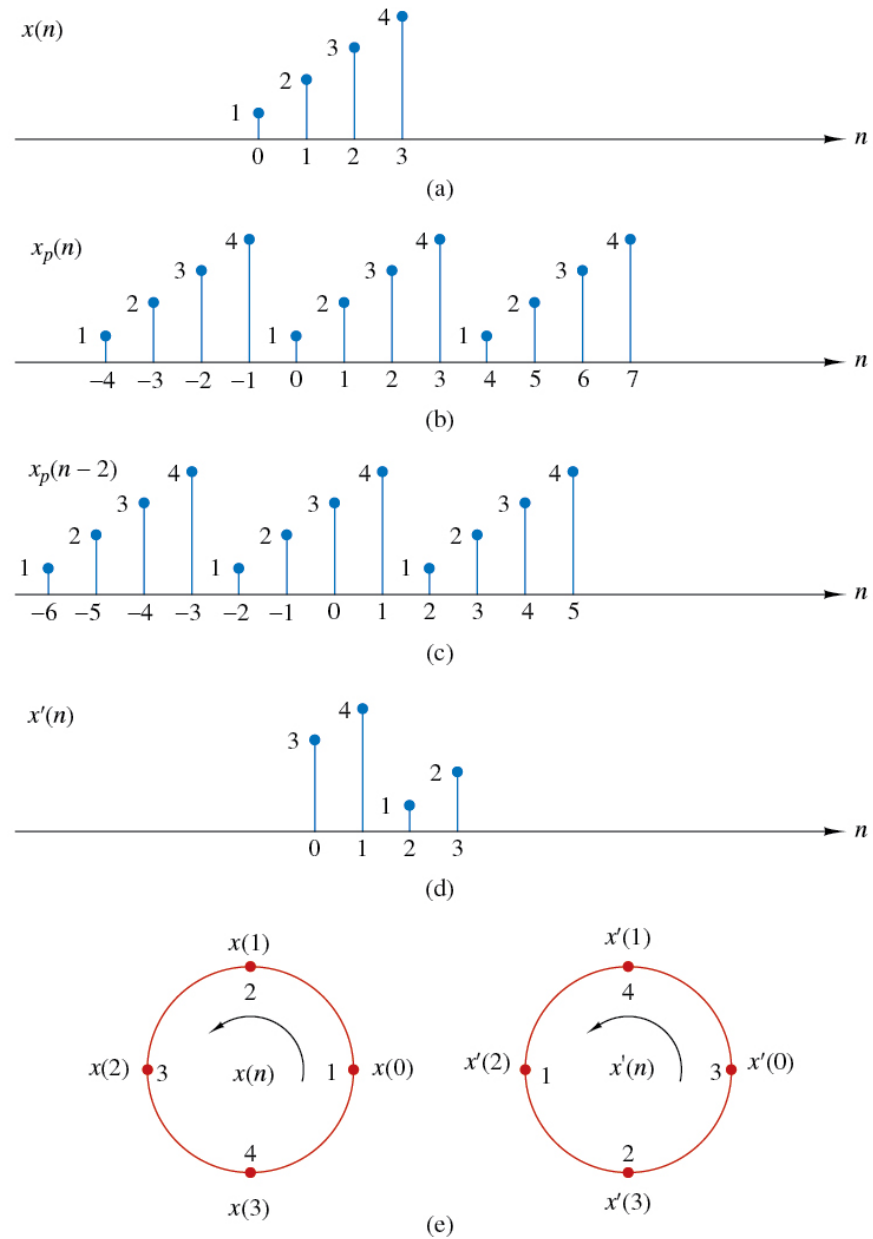


Figure 7.2.1 Circular shift of a sequence.



Equation 7.2.31

$$\begin{array}{c} x(n) = x_R^e(n) + x_R^o(n) + jx_I^e(n) + jx_I^o(n) \\ \updownarrow \quad \swarrow \quad \downarrow \quad \searrow \\ X(k) = X_R^e(k) + X_R^o(k) + jX_I^e(k) + jX_I^o(k) \end{array}$$

Table 7.1 Symmetry Properties of the DFT

N -Point Sequence $x(n)$, $0 \leq n \leq N - 1$	N -Point DFT
$x(n)$	$X(k)$
$x^*(n)$	$X^*(N - k)$
$x^*(N - n)$	$X^*(k)$
$x_R(n)$	$X_{ce}(k) = \frac{1}{2}[X(k) + X^*(N - k)]$
$jX_I(n)$	$X_{co}(k) = \frac{1}{2}[X(k) - X^*(N - k)]$
$x_{ce}(n) = \frac{1}{2}[x(n) + x^*(N - n)]$	$X_R(k)$
$x_{co}(n) = \frac{1}{2}[x(n) - x^*(N - n)]$	$jX_I(k)$
Real Signals	
Any real signal	$X(k) = X^*(N - k)$
$x(n)$	$X_R(k) = X_R(N - k)$
	$X_I(k) = -X_I(N - k)$
	$ X(k) = X(N - k) $
	$\angle X(k) = -\angle X(N - k)$
$x_{ce}(n) = \frac{1}{2}[x(n) + x(N - n)]$	$X_R(k)$
$x_{co}(n) = \frac{1}{2}[x(n) - x(N - n)]$	$jX_I(k)$

Figure 7.2.2 Circular convolution of two sequences.

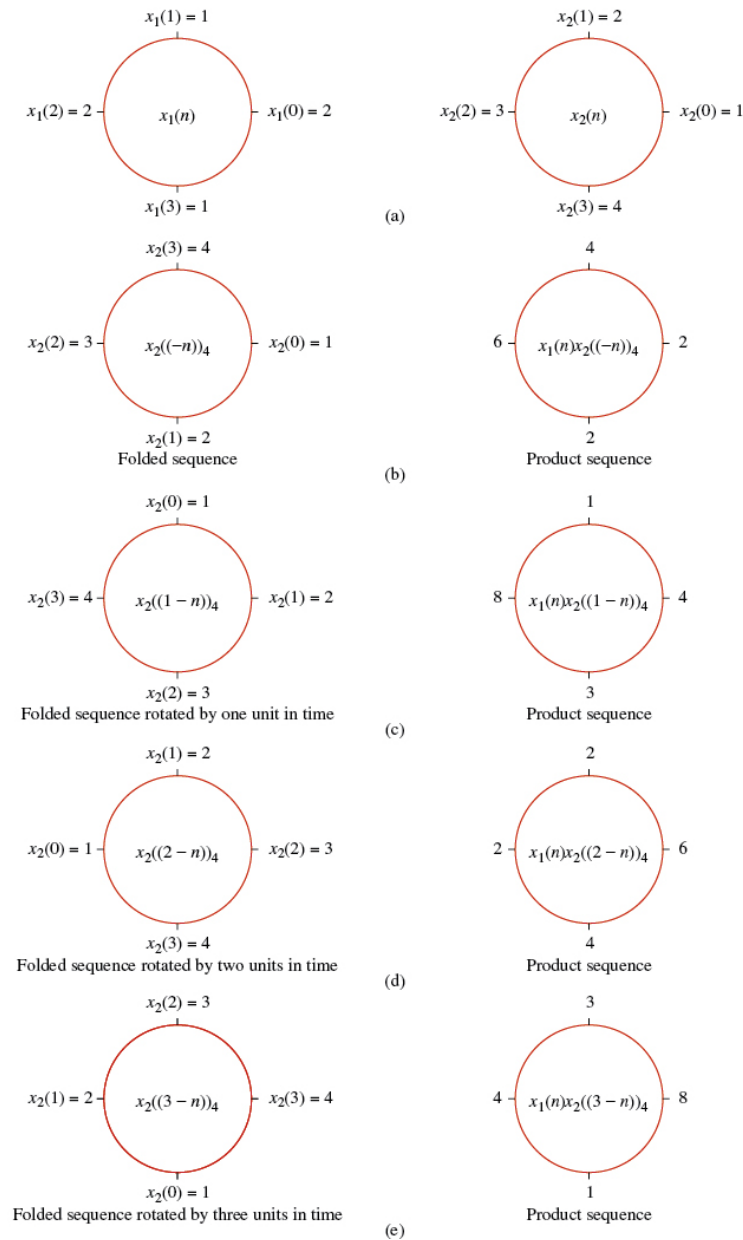


Figure 7.2.3 Time reversal of a sequence.

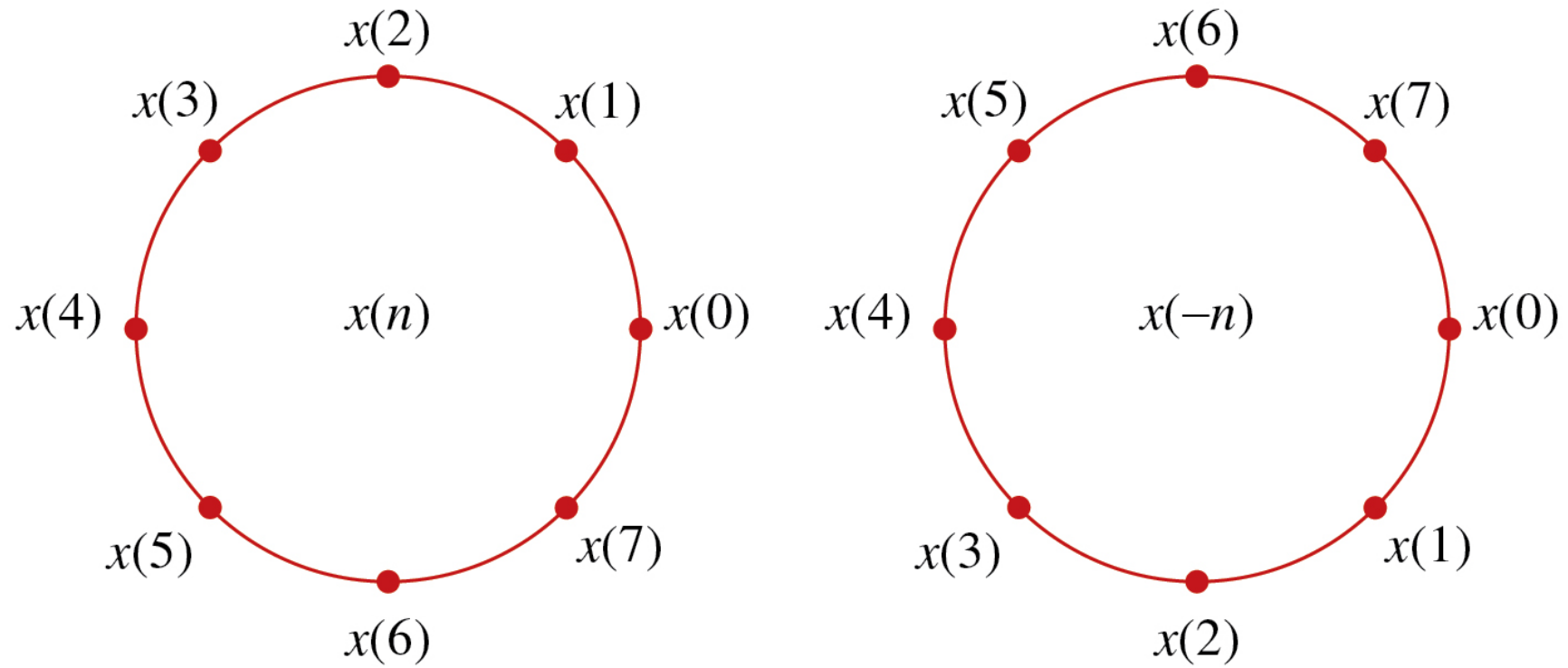


Table 7.2 Properties of the DFT

Property	Time Domain	Frequency Domain
Notation	$x(n), y(n)$	$X(k), Y(k)$
Periodicity	$x(n) = x(n + N)$	$X(k) = X(k + N)$
Linearity	$a_1x_1(n) + a_2x_2(n)$	$a_1X_1(k) + a_2X_2(k)$
Time reversal	$x(N - n)$	$X(N - k)$
Circular time shift	$x((n - l))_N$	$X(k)e^{-j2\pi kl/N}$
Circular frequency shift	$x(n)e^{j2\pi ln/N}$	$X((k - l))_N$
Complex conjugate	$x^*(n)$	$X^*(N - k)$
Circular convolution	$x_1(n) \circledast x_2(n)$	$X_1(k)X_2(k)$
Circular correlation	$x(n) \circledast y^*(-n)$	$X(k)Y^*(k)$
Multiplication of two sequences	$x_1(n)x_2(n)$	$\frac{1}{N}X_1(k) \circledast X_2(k)$
Parseval's theorem	$\sum_{n=0}^{N-1} x(n)y^*(n)$	$\frac{1}{N} \sum_{k=0}^{N-1} X(k)Y^*(k)$

Figure 7.3.1 Linear FIR filtering by the overlap-save method.

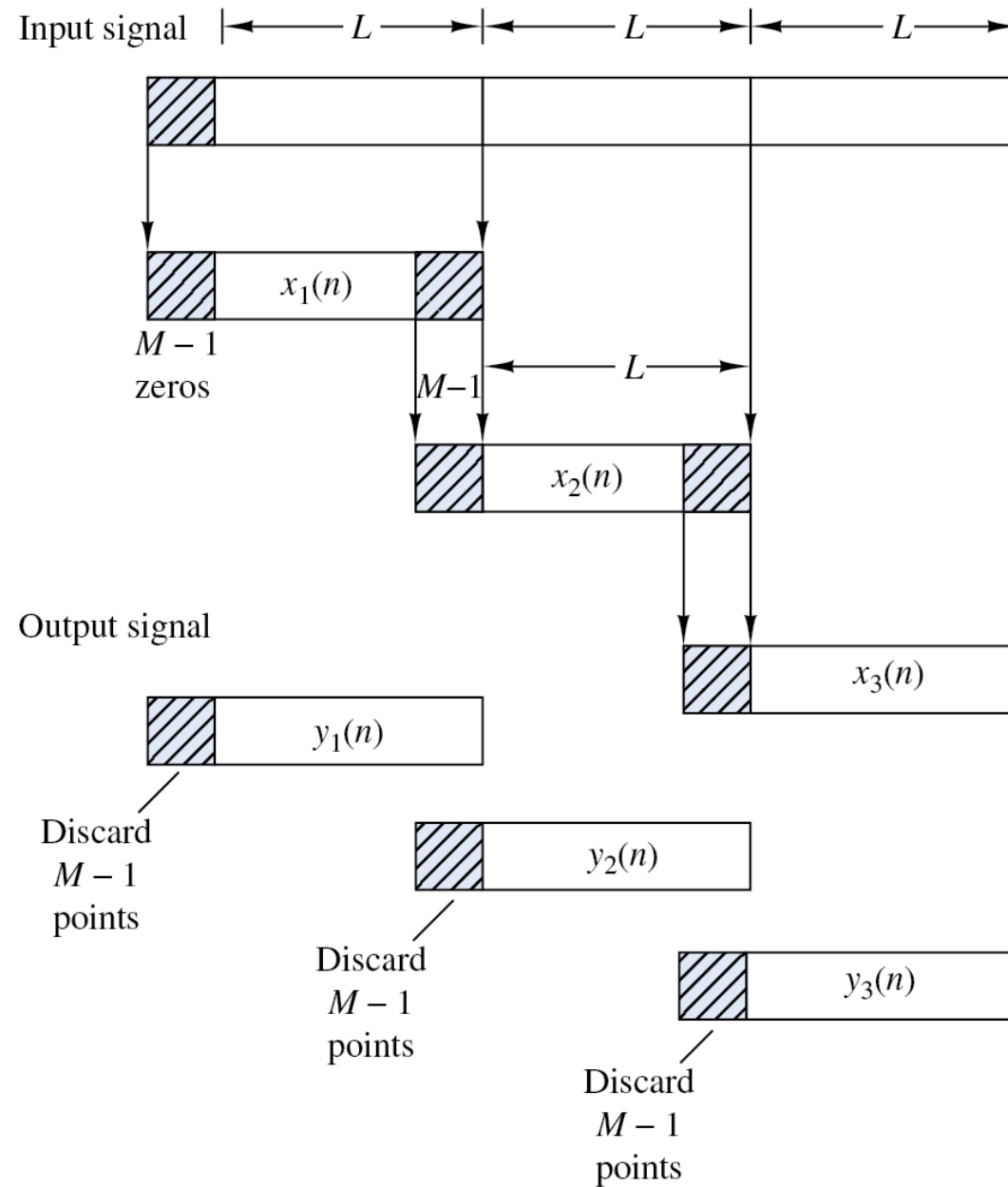


Figure 7.3.2 Linear FIR filtering by the overlap-add method.

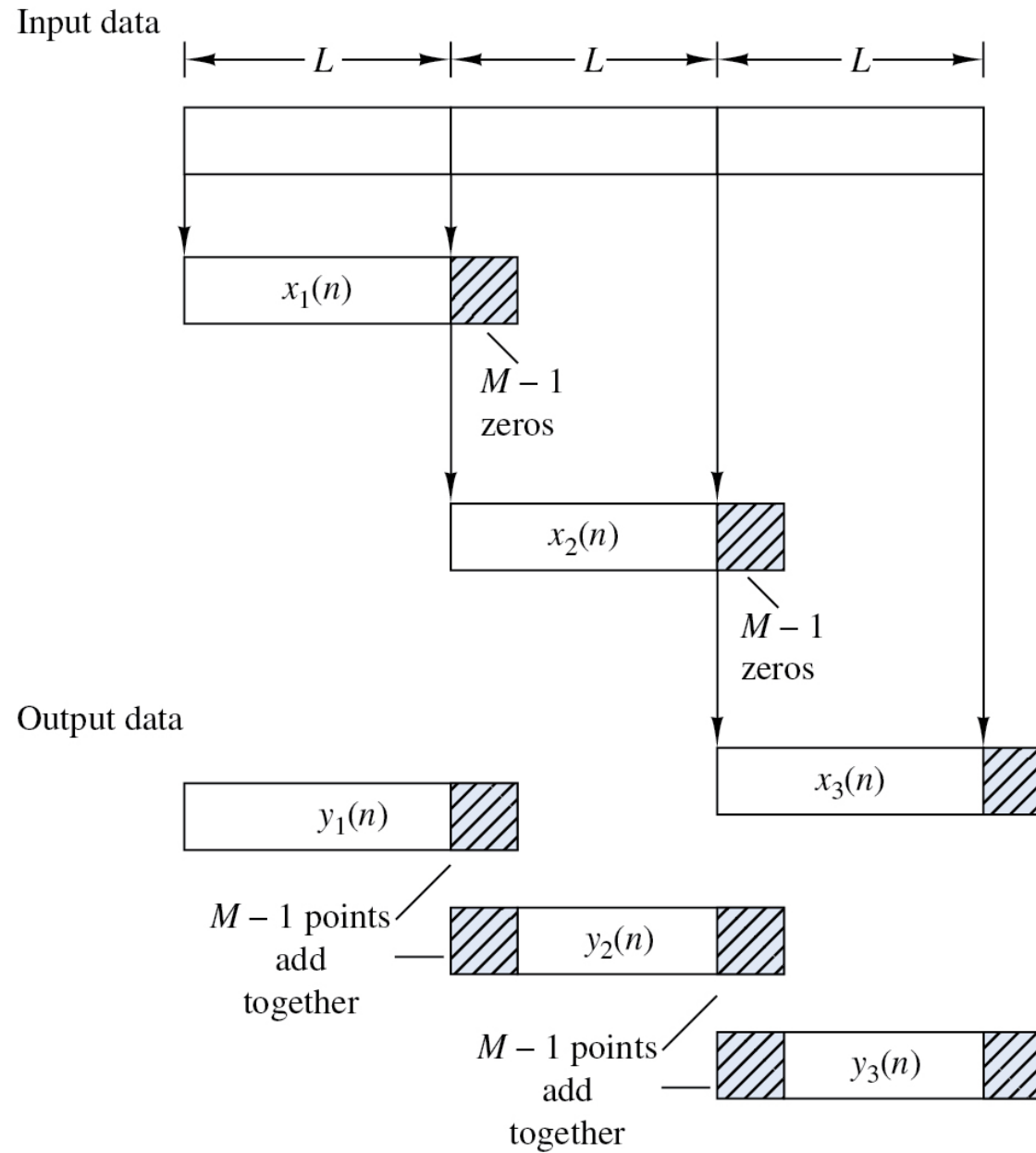


Figure 7.4.1 Magnitude spectrum for $L = 25$ and $N = 2048$, illustrating the occurrence of leakage.

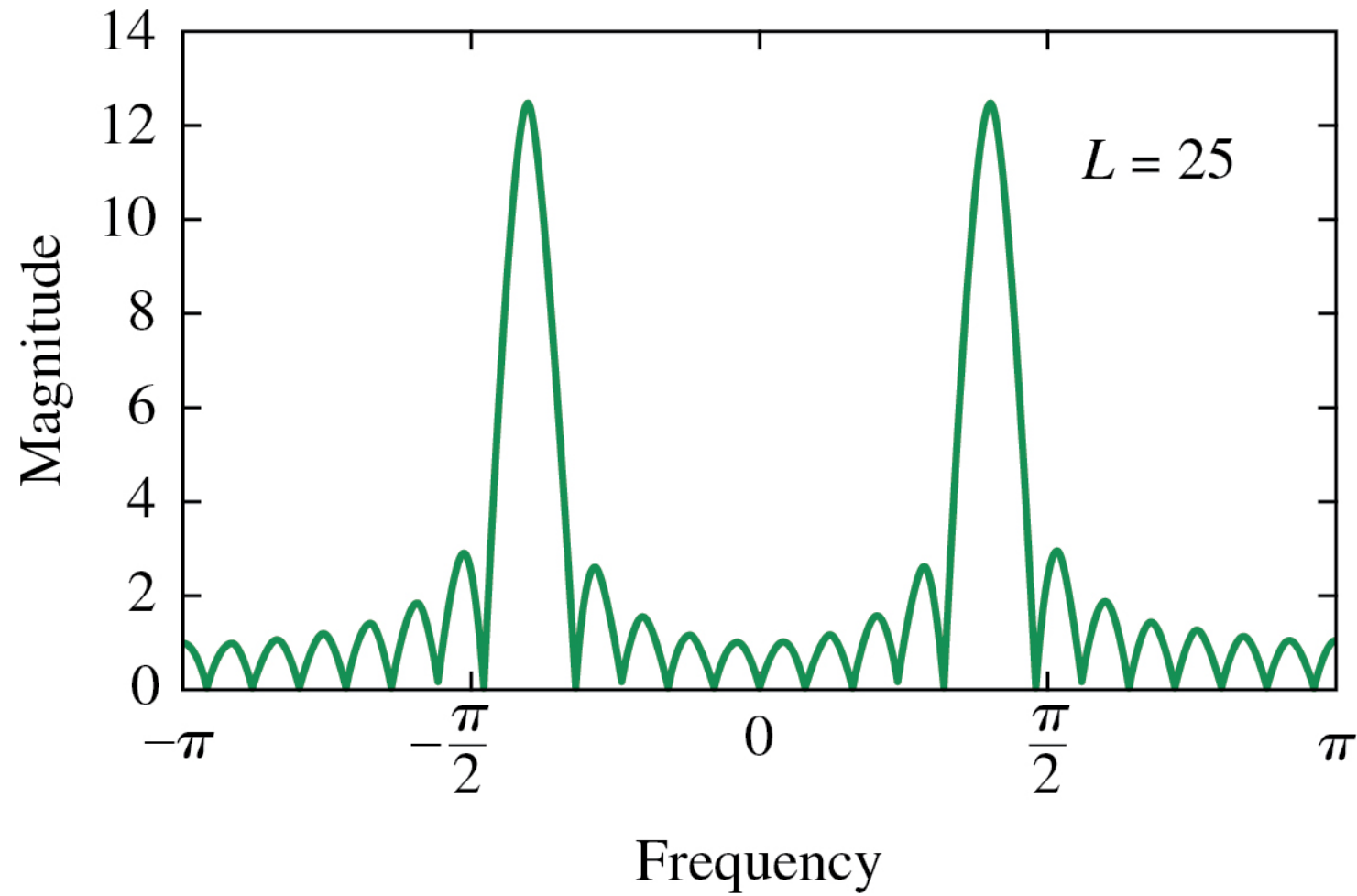
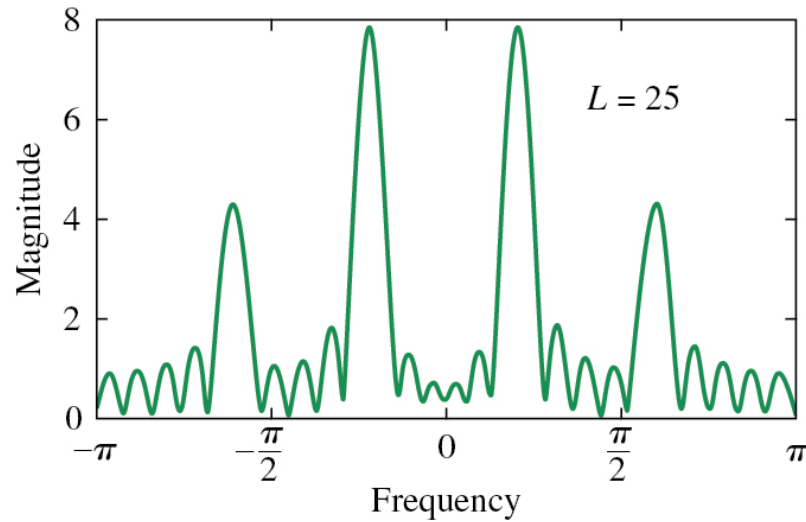
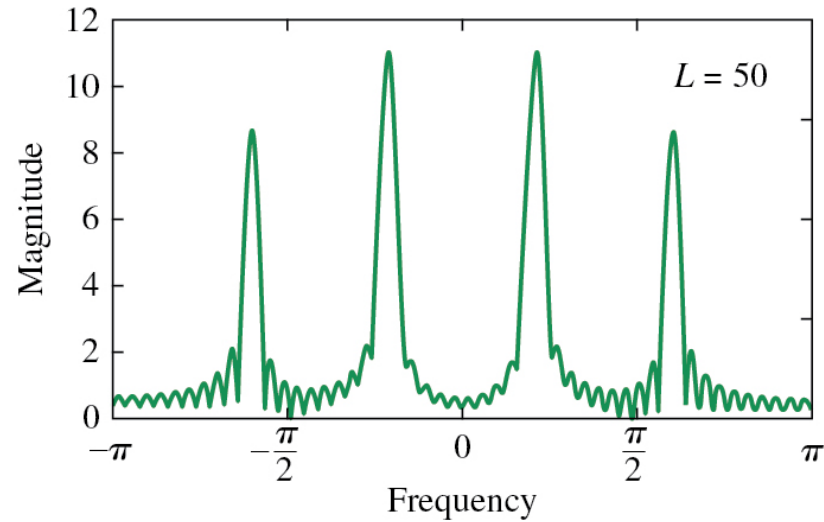


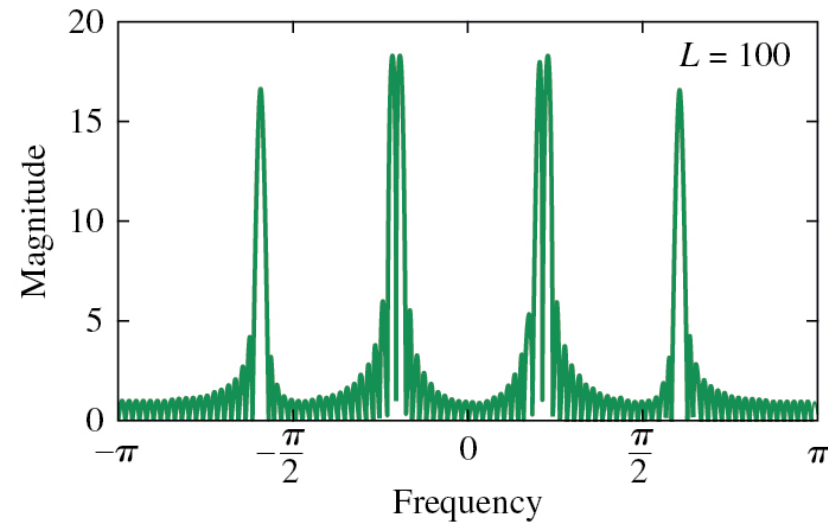
Figure 7.4.2 Magnitude spectrum for the signal given by (7.4.8), as observed through a rectangular window.



(a)



(b)



(c)

Figure 7.4.3 Magnitude spectrum of the Hanning window.

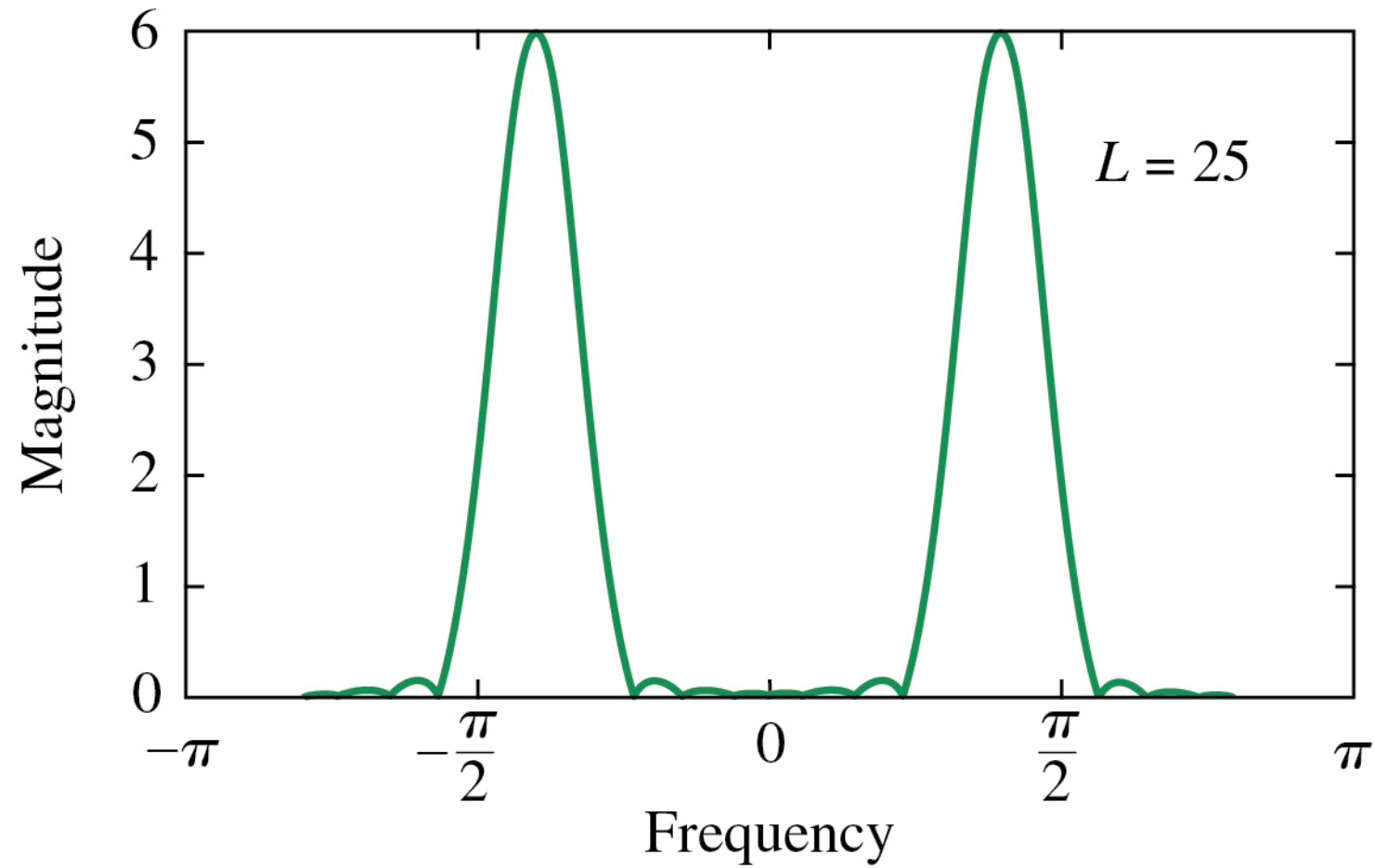


Figure 7.4.4 Magnitude spectrum of the signal in (7.4.8) as observed through a Hanning window.

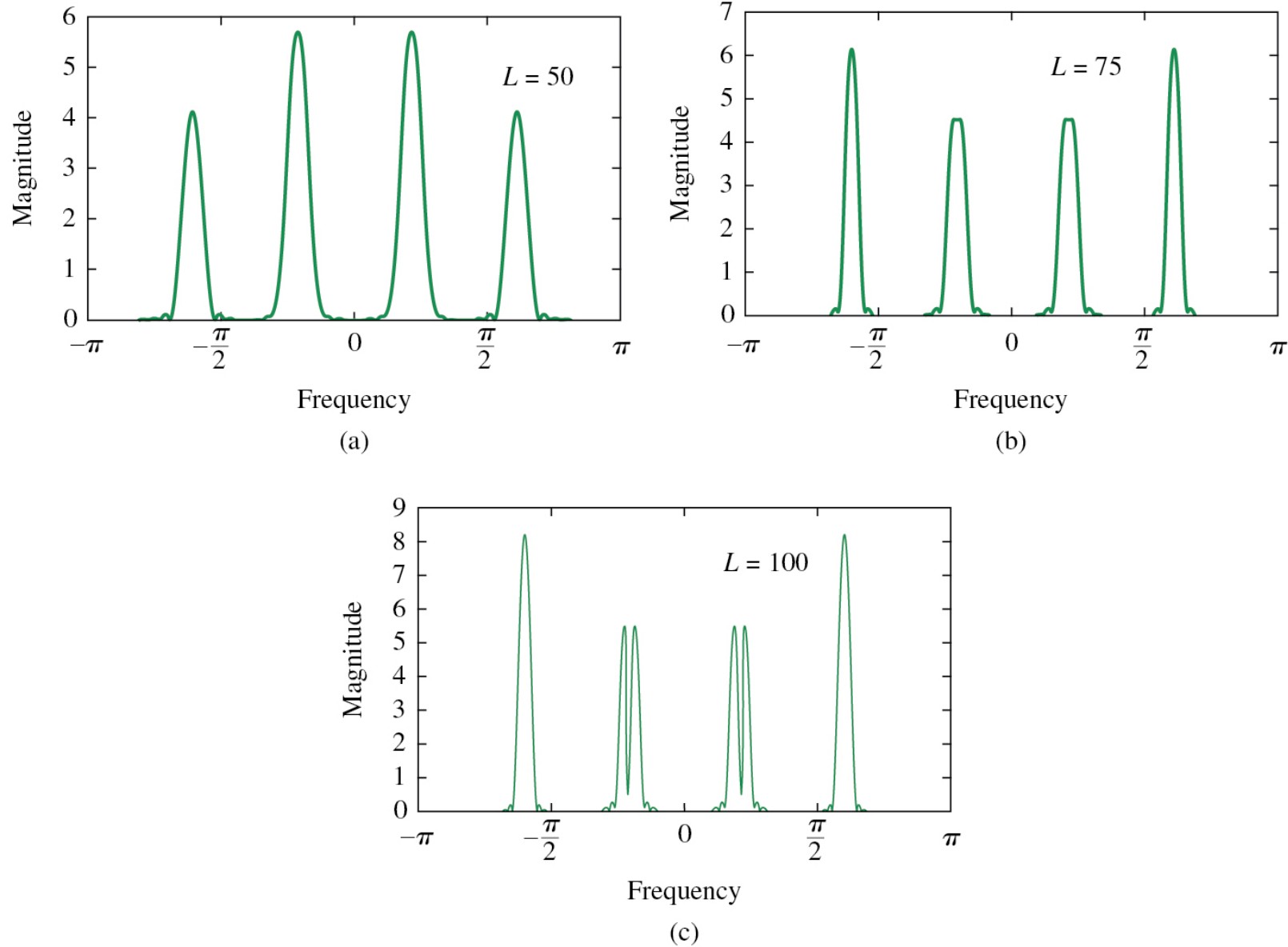
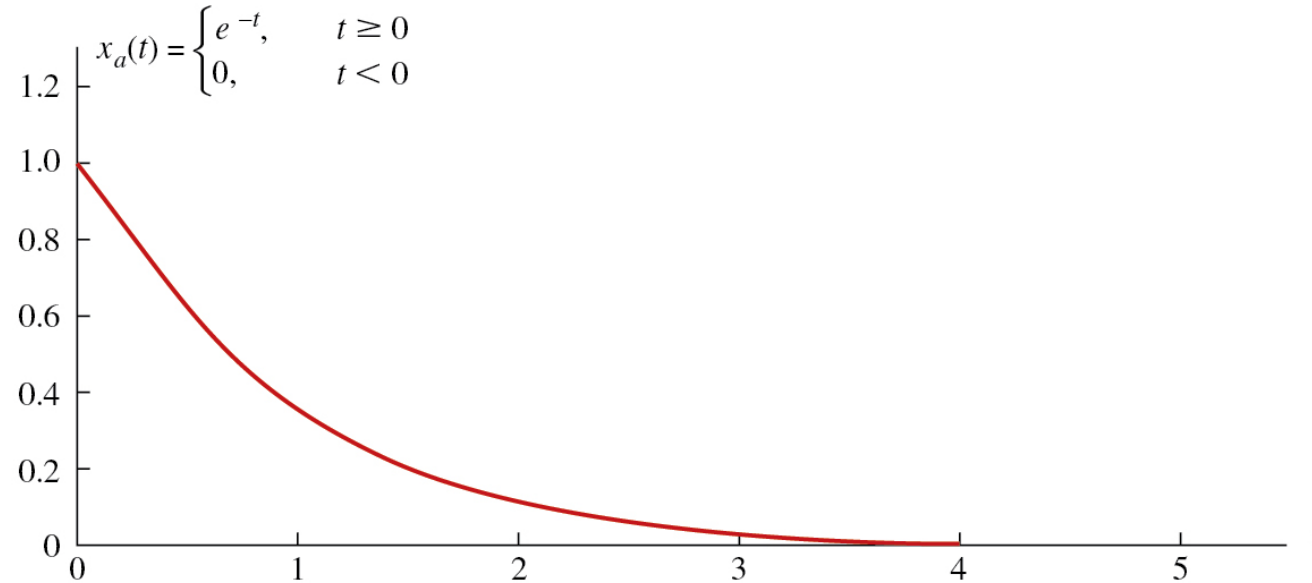
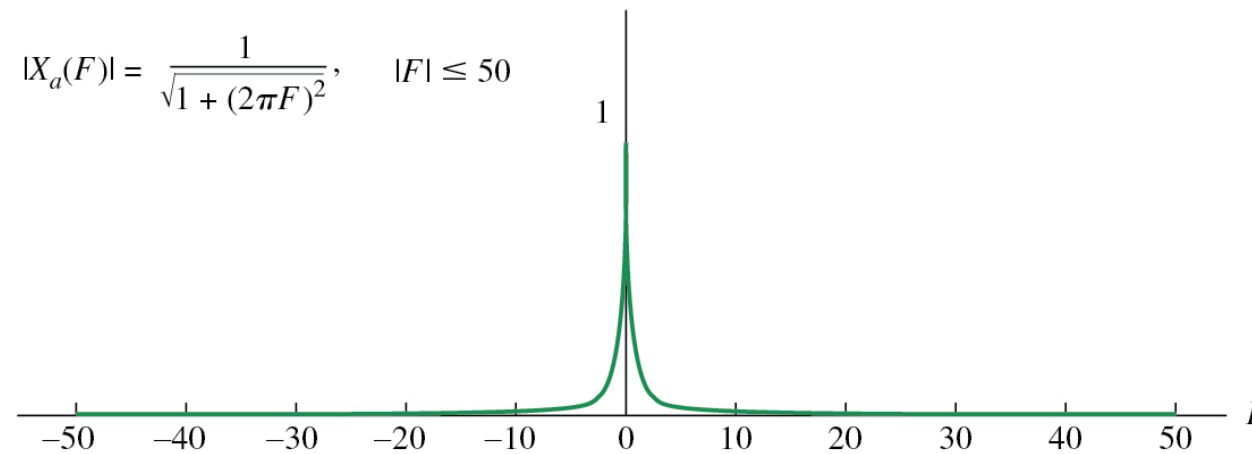


Figure 7.4.5 Effect of windowing (truncating) the sampled version of the analog signal in Example 7.4.1.

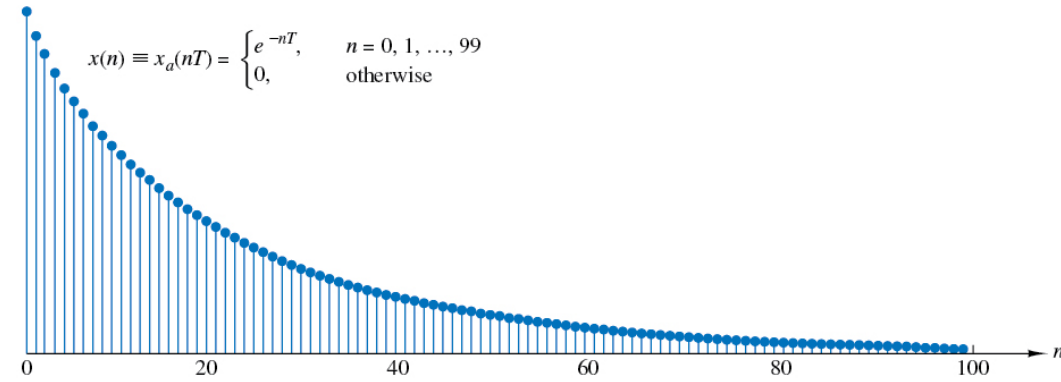


(a)

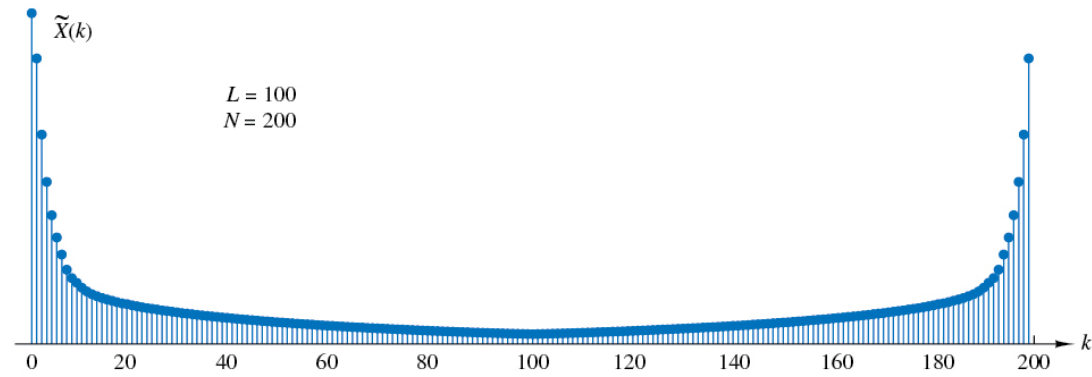


(b)

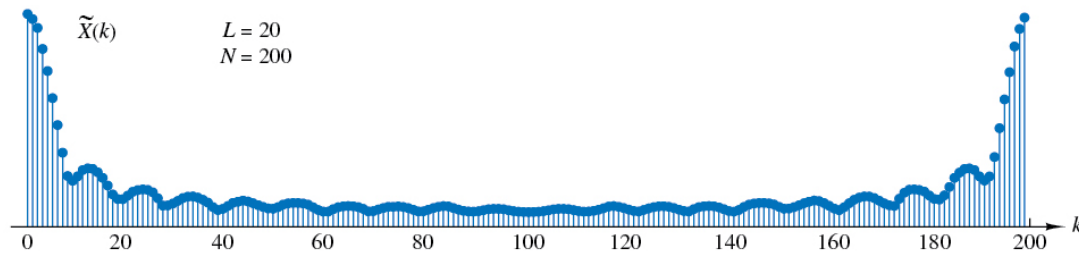
Figure 7.4.5 (Continued)



(c)



(d)



(e)

Figure 7.4.6 Illustration of bandlimited interpolation using the DFT.

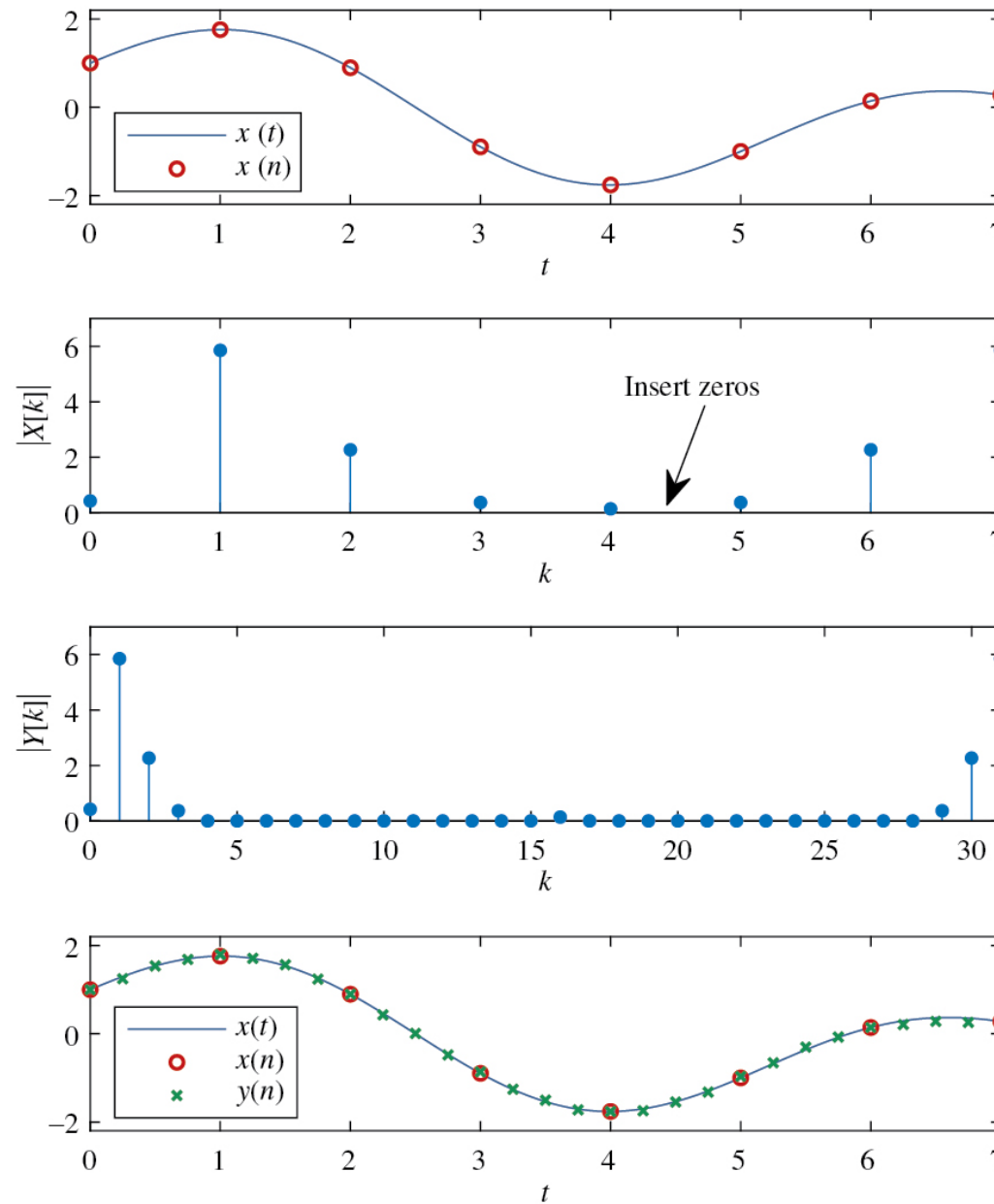


Figure 7.5.1 The STFT for the sinusoidal signal defined by equation (7.5.11).

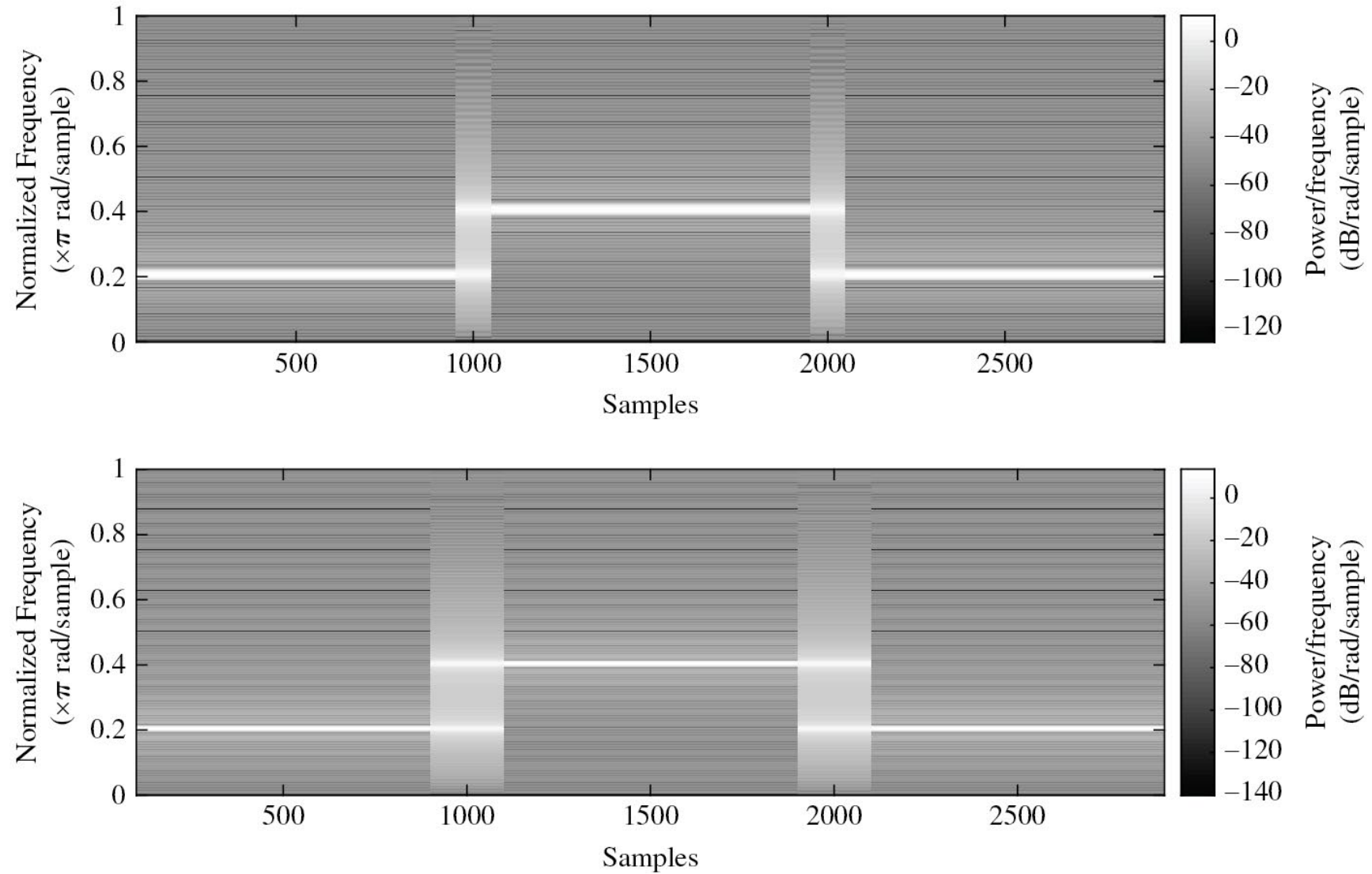
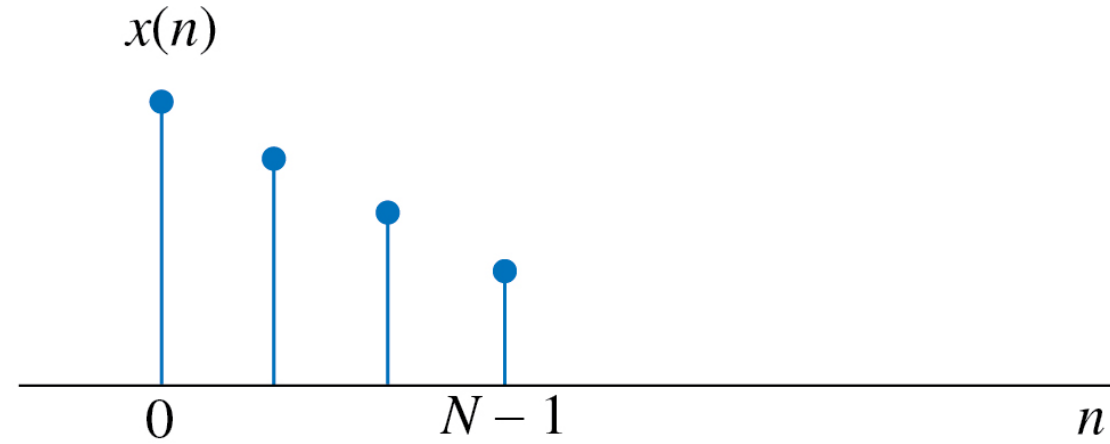
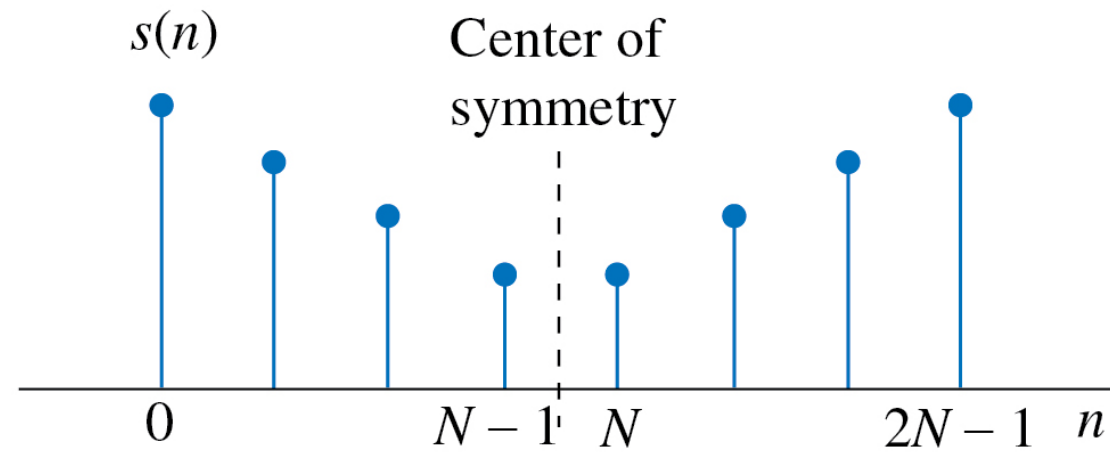


Figure 7.6.1 Original sequence $x(n)$, $0 \leq n \leq N - 1$ and its $2N$ -point even extension $s(n)$, $0 \leq n \leq 2N - 1$.



(a)



(b)

Figure 7.6.2 A discrete-time sinusoidal signal and its DFT and DCT representations.

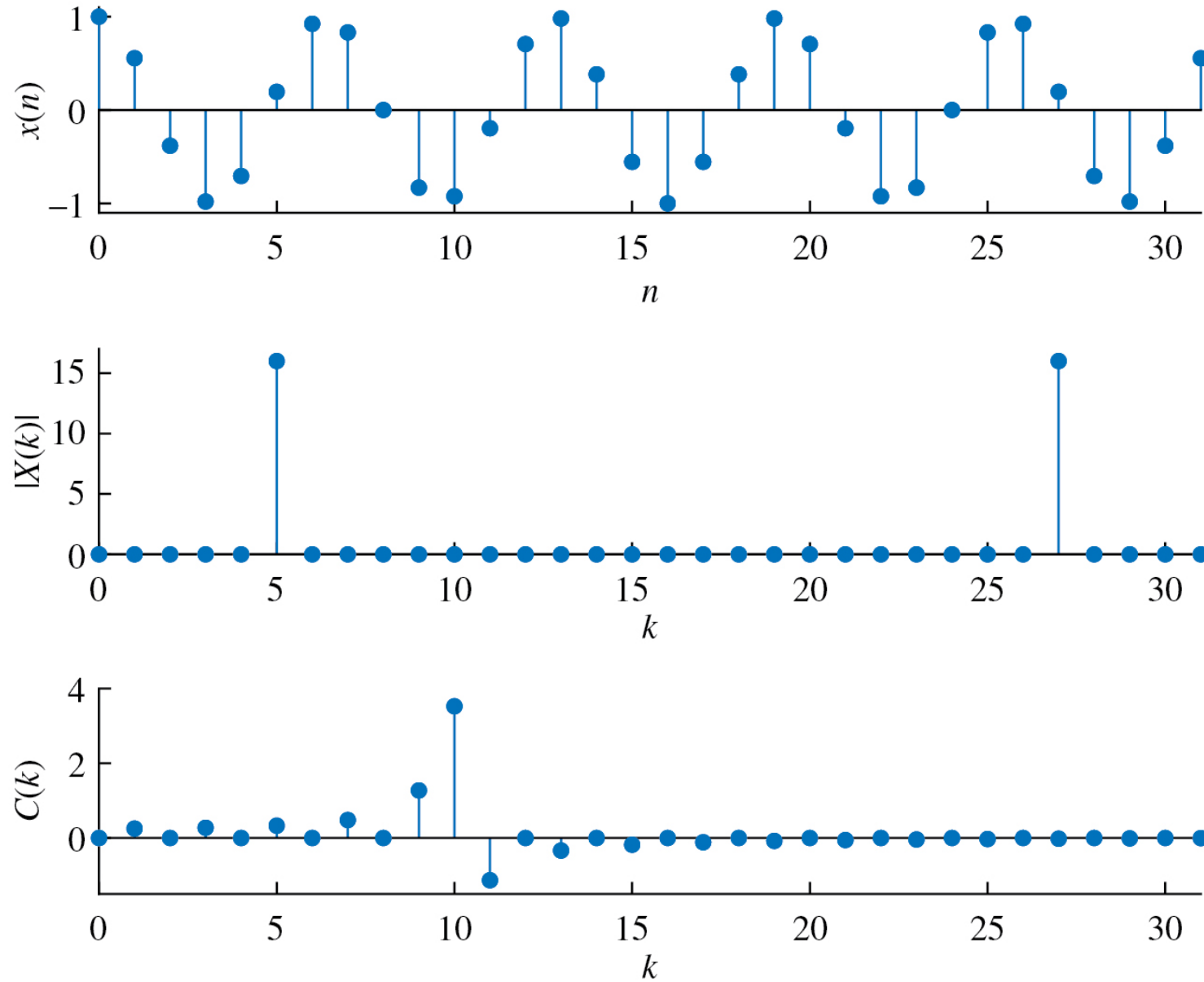


Figure 7.6.3 A discrete time sinusoidal signal and its DFT and DCT representations.

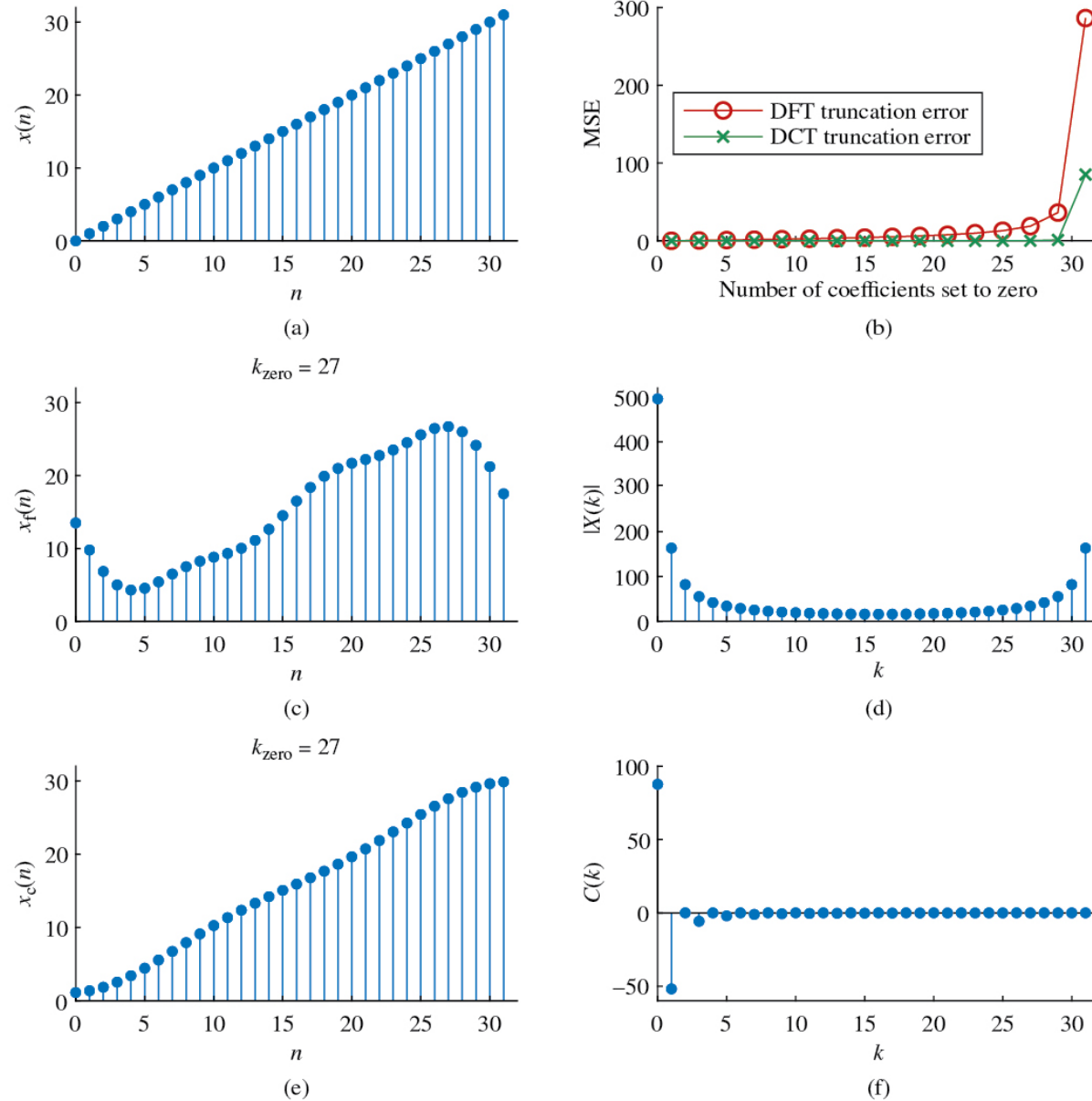


Figure CP7.8

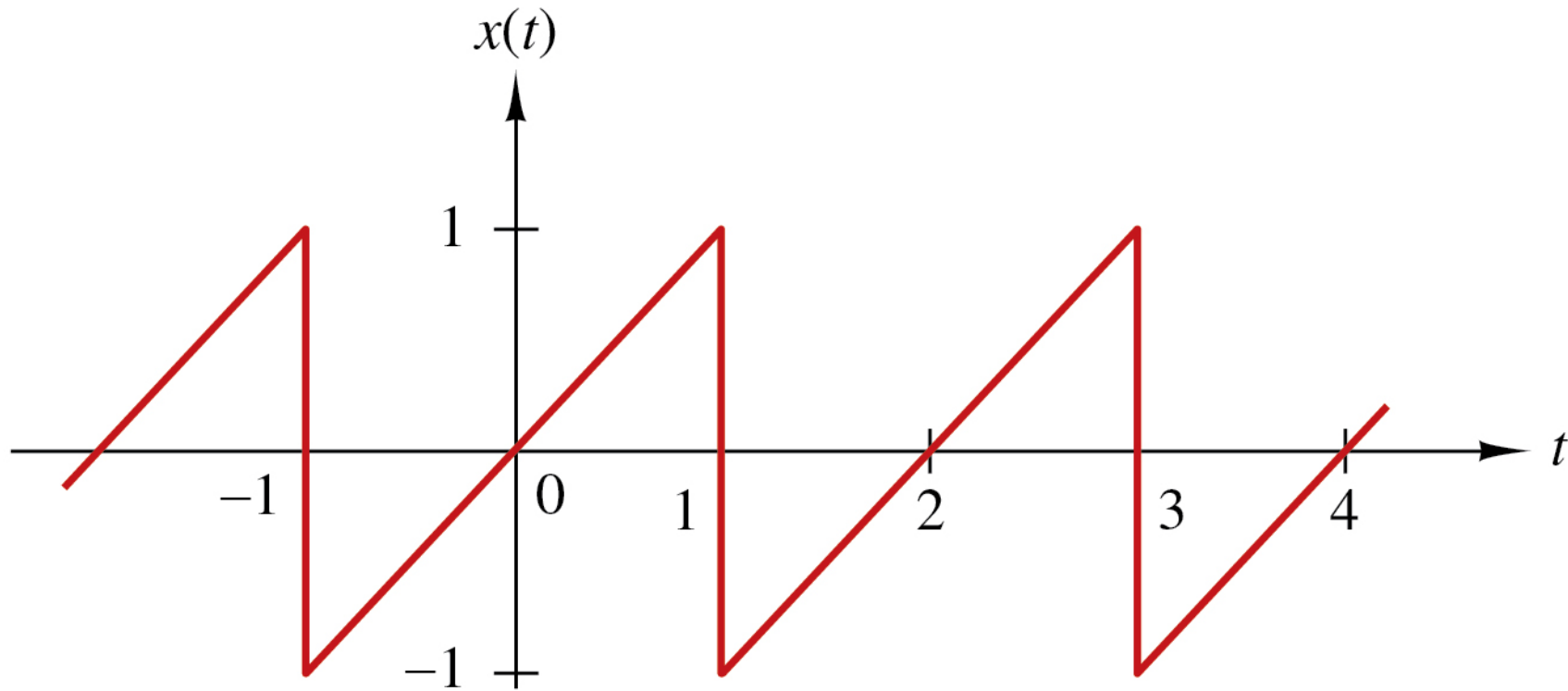
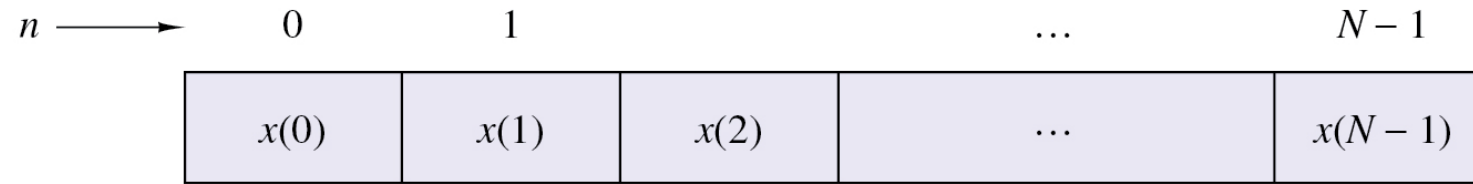
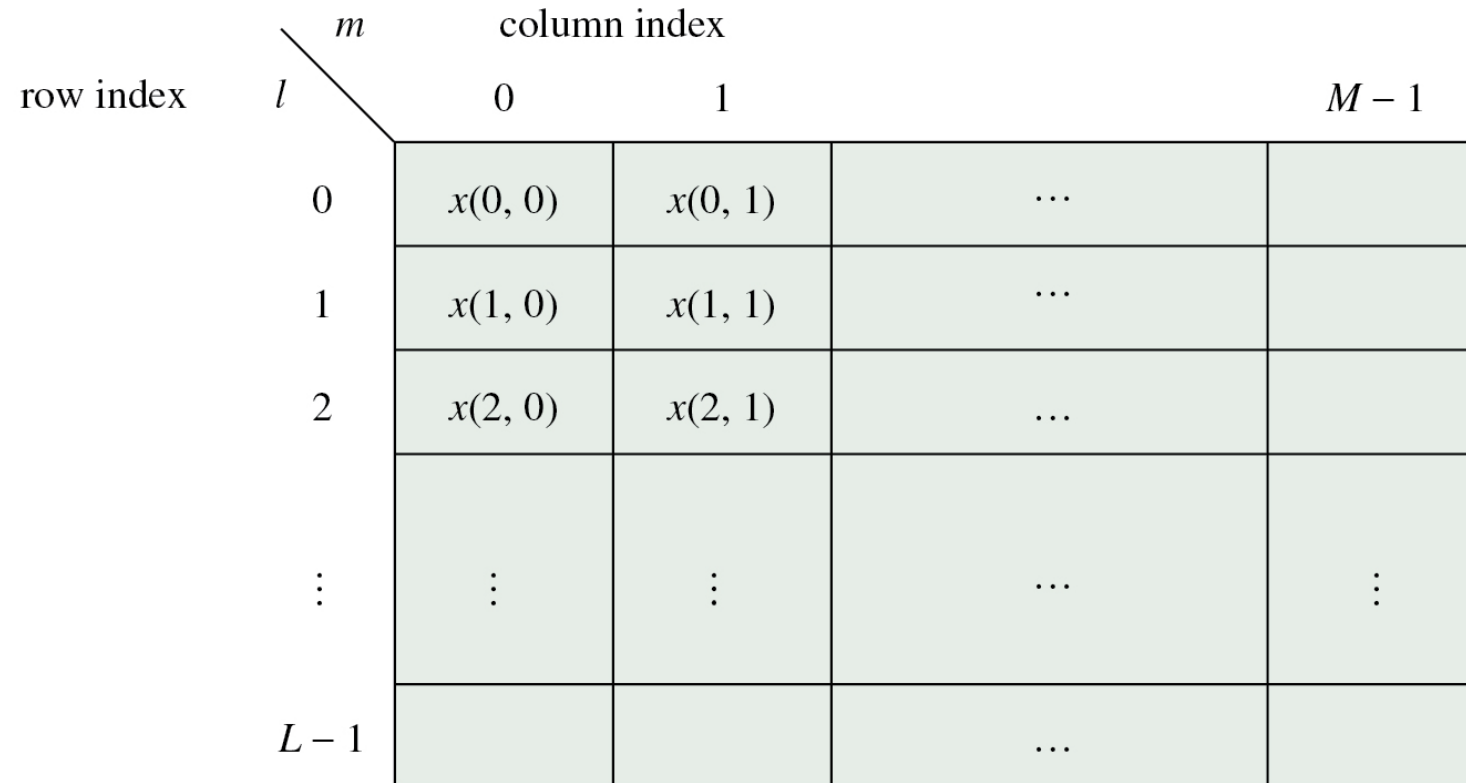


Figure 8.1.1 Two dimensional data array for storing the sequence $x(n)$, $0 \leq n \leq N - 1$.



(a)



(b)

Figure 8.1.2 Two arrangements for the data arrays.

Row-wise $n = Ml + m$

$l \backslash m$	0	1	2	...	$M - 1$
0	$x(0)$	$x(1)$	$x(2)$	...	$x(M - 1)$
1	$x(M)$	$x(M + 1)$	$x(M + 2)$	...	$x(2M - 1)$
2	$x(2M)$	$x(2M + 1)$	$x(2M + 2)$	...	$x(3M - 1)$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	...	$\vdots$
$L - 1$	$x((L - 1)M)$	$x((L - 1)M + 1)$	$x((L - 1)M + 2)$	...	$x(LM - 1)$

(a)

Column-wise $n = l + mL$

$l \backslash m$	0	1	2	...	$M - 1$
0	$x(0)$	$x(L)$	$x(2L)$	...	$x((M - 1)L)$
1	$x(1)$	$x(L + 1)$	$x(2L + 1)$	...	$x((M - 1)L + 1)$
2	$x(2)$	$x(L + 2)$	$x(2L + 2)$	...	$x((M - 1)L + 2)$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	...	$\vdots$
$L - 1$	$x(L - 1)$	$x(2L - 1)$	$x(3L - 1)$	...	$x(LM - 1)$

(b)

Figure 8.1.3 Computation of $N = 15$ -point DFT by means of 3-point and 5-point DFTs.

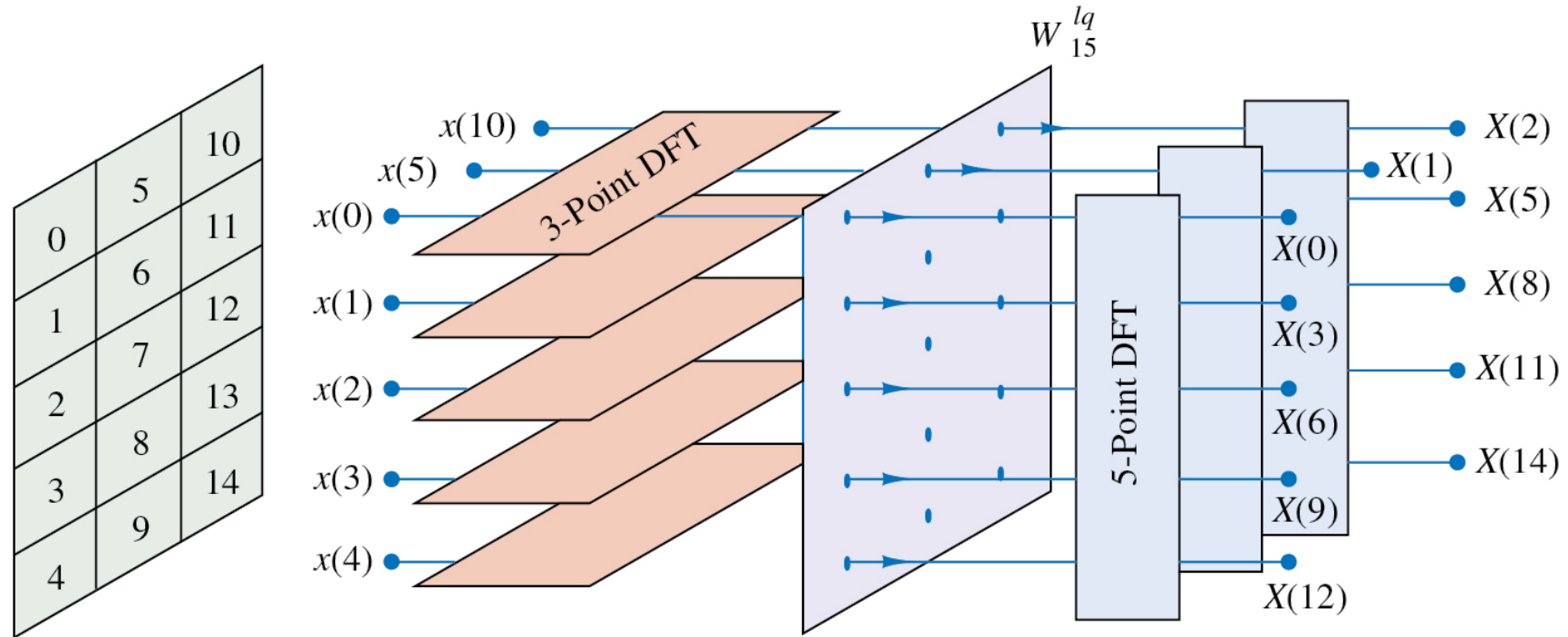


Figure 8.1.4 First step in the decimation-in-time algorithm.

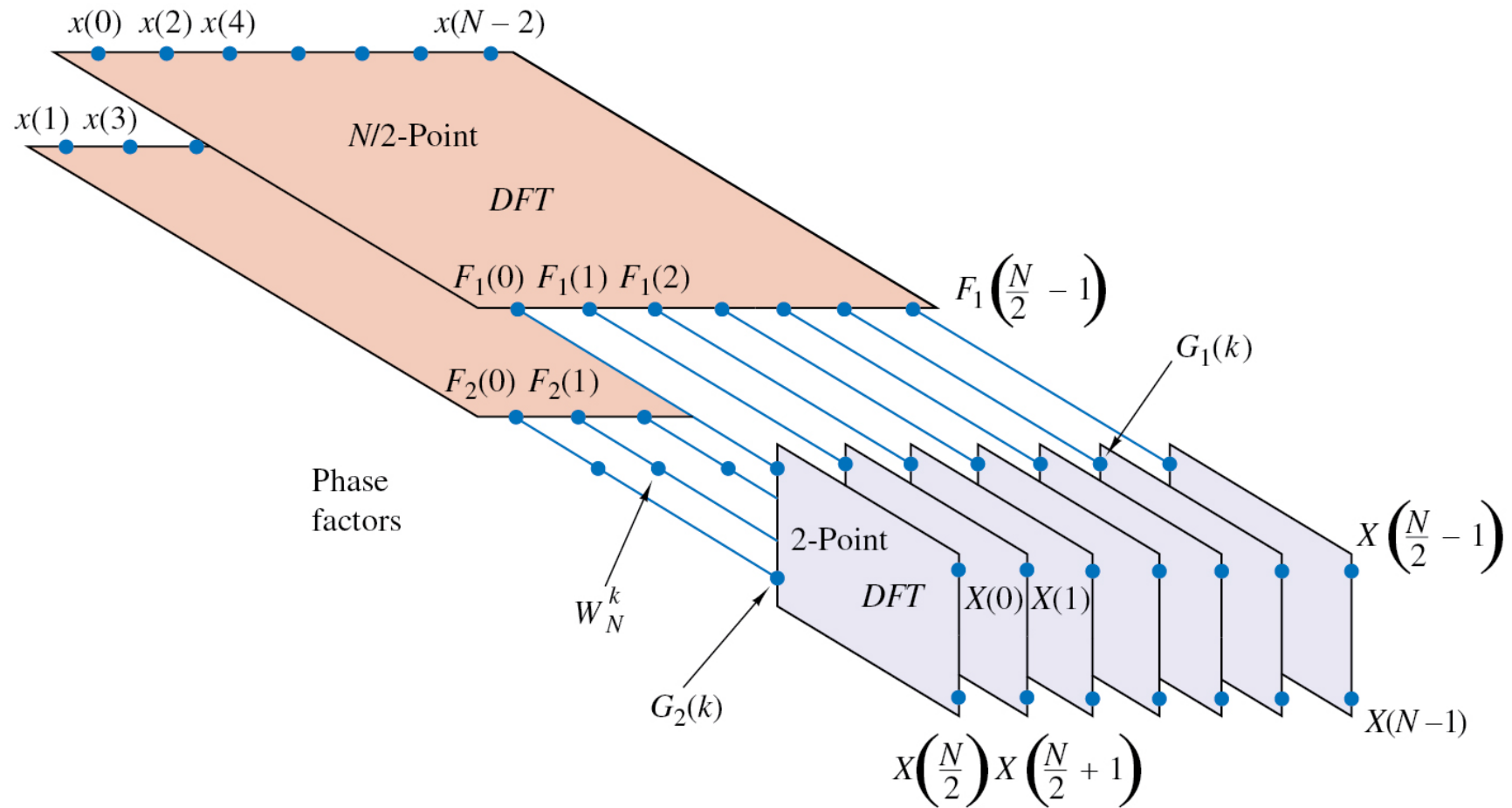


Table 8.1 Comparison of Computational Complexity for the Direct Computation of the DFT Versus the FFT Algorithm

Number of Points, N	Complex Multiplications in Direct Computation, N^2	Complex Multiplications in FFT Algorithm, $(N/2) \log_2 N$	Speed Improvement Factor
4	16	4	4.0
8	64	12	5.3
16	256	32	8.0
32	1,024	80	12.8
64	4,096	192	21.3
128	16,384	448	36.6
256	65,536	1,024	64.0
512	262,144	2,304	113.8
1,024	1,048,576	5,120	204.8

Figure 8.1.5 Three stages in the computation of an $N = 8$ -point DFT.

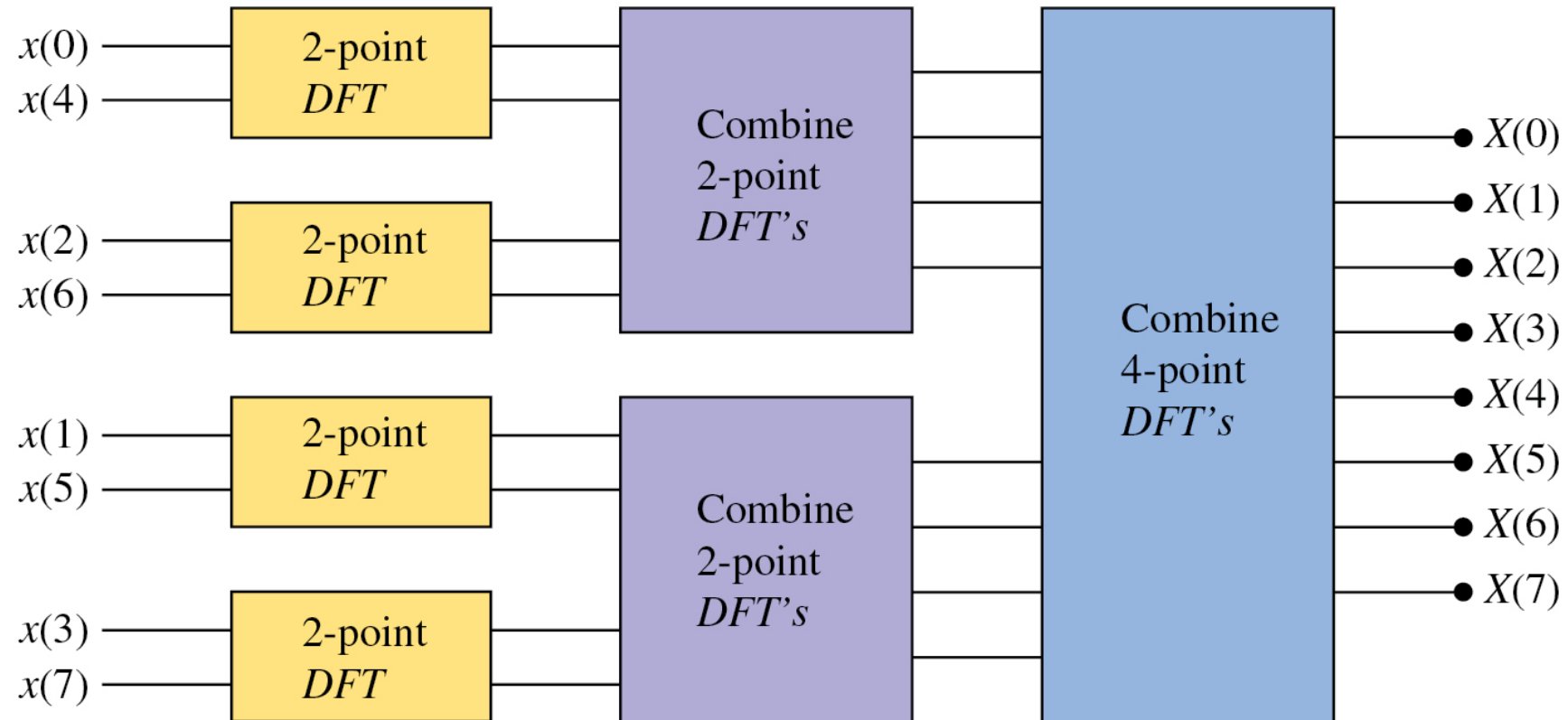


Figure 8.1.6 Eight-point decimation-in-time FFT algorithm.

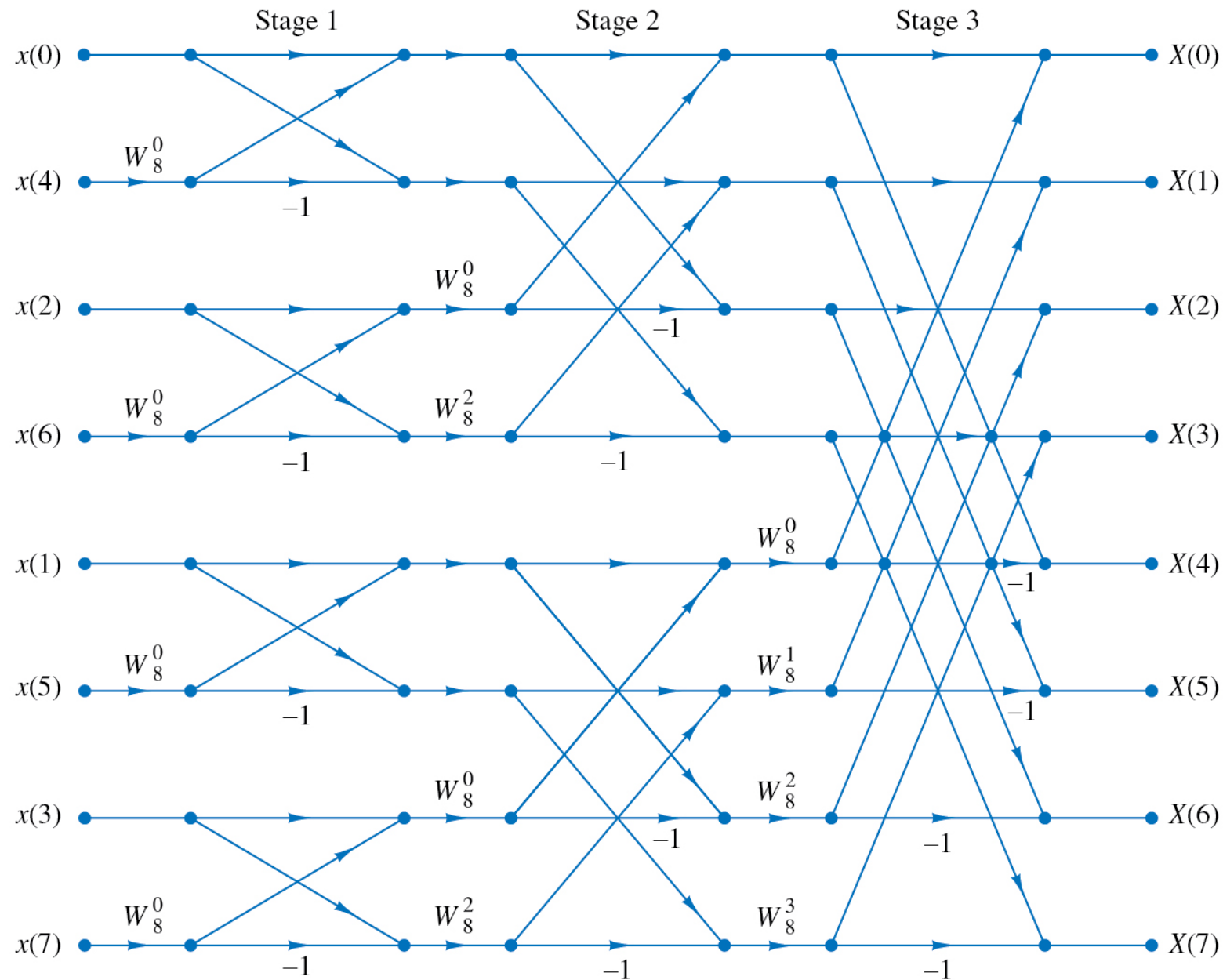


Figure 8.1.7 Basic butterfly computation in the decimation-in-time FFT algorithm.

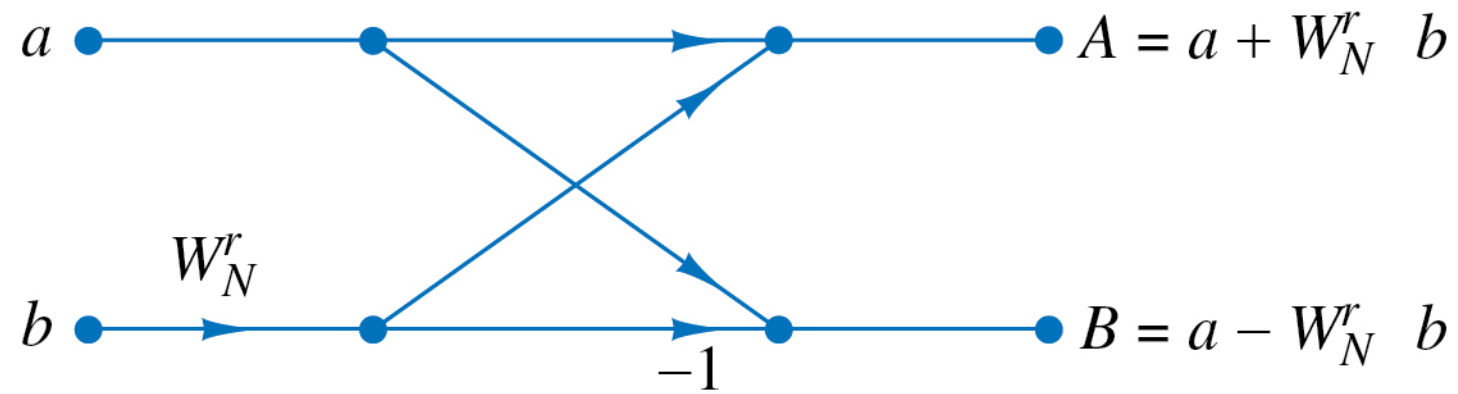
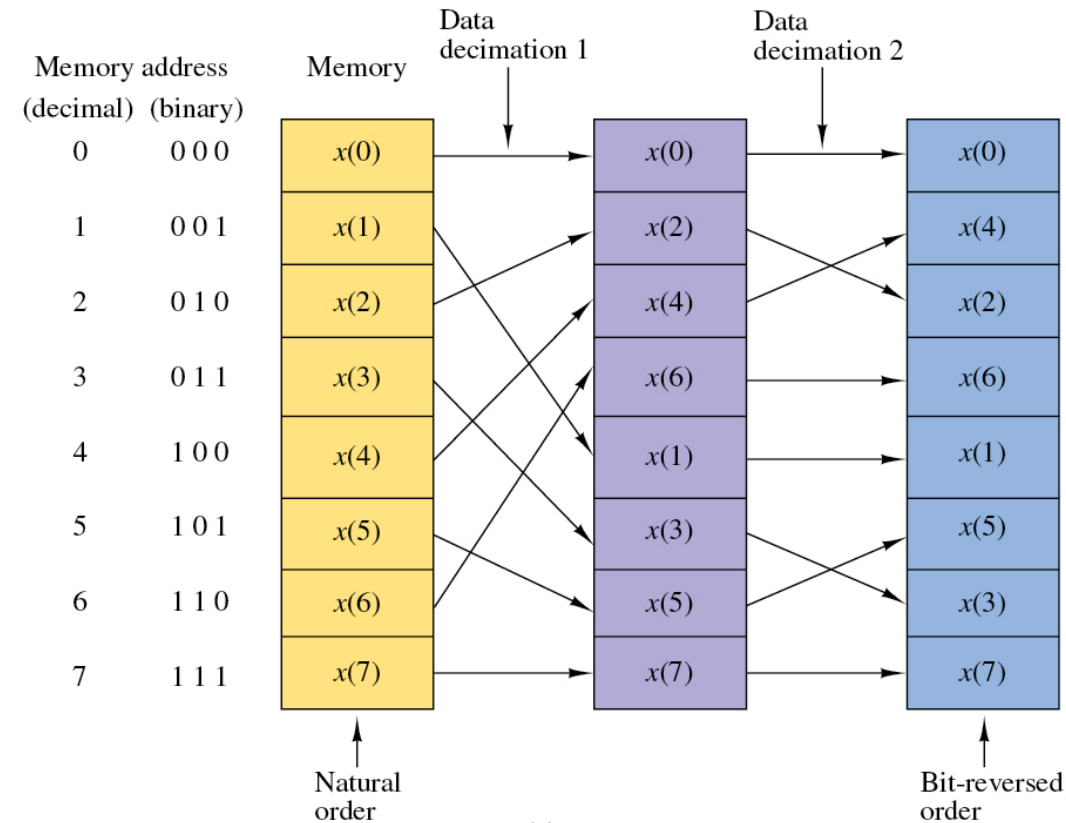


Figure 8.1.8 Shuffling of the data and bit reversal.



(a)

$(n_2 n_1 n_0)$	$\rightarrow$	$(n_1 n_0 n_2)$	$\rightarrow$	$(n_0 n_1 n_2)$
(0 0 0)	$\rightarrow$	(0 0 0)	$\rightarrow$	(0 0 0)
(0 0 1)	$\rightarrow$	(0 1 0)	$\rightarrow$	(1 0 0)
(0 1 0)	$\rightarrow$	(1 0 0)	$\rightarrow$	(0 1 0)
(0 1 1)	$\rightarrow$	(1 1 0)	$\rightarrow$	(1 1 0)
(1 0 0)	$\rightarrow$	(0 0 1)	$\rightarrow$	(0 0 1)
(1 0 1)	$\rightarrow$	(0 1 1)	$\rightarrow$	(1 0 1)
(1 1 0)	$\rightarrow$	(1 0 1)	$\rightarrow$	(0 1 1)
(1 1 1)	$\rightarrow$	(1 1 1)	$\rightarrow$	(1 1 1)

(b)

Figure 8.1.9 First stage of the decimation-in-frequency FFT algorithm.

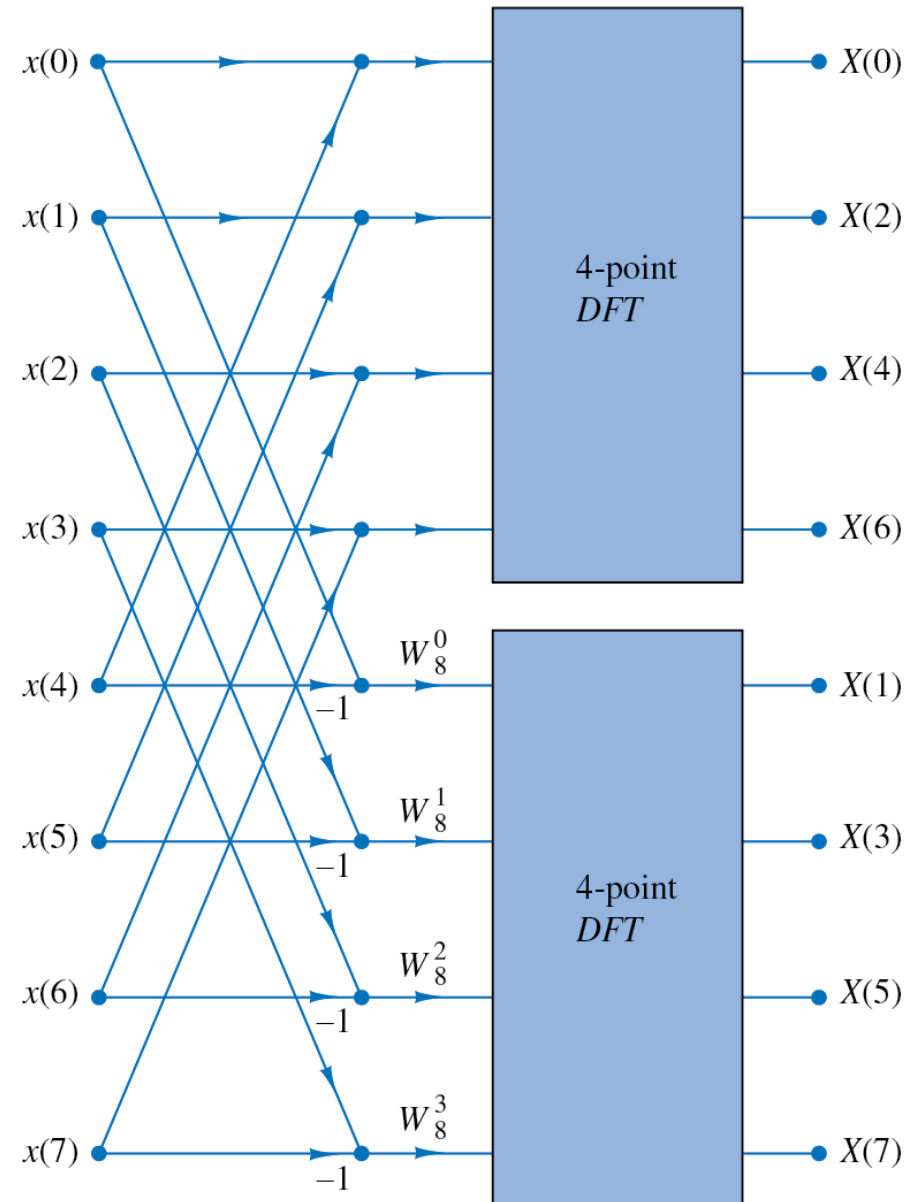


Figure 8.1.10 Basic butterfly computation in the decimation-in-frequency FFT algorithm.

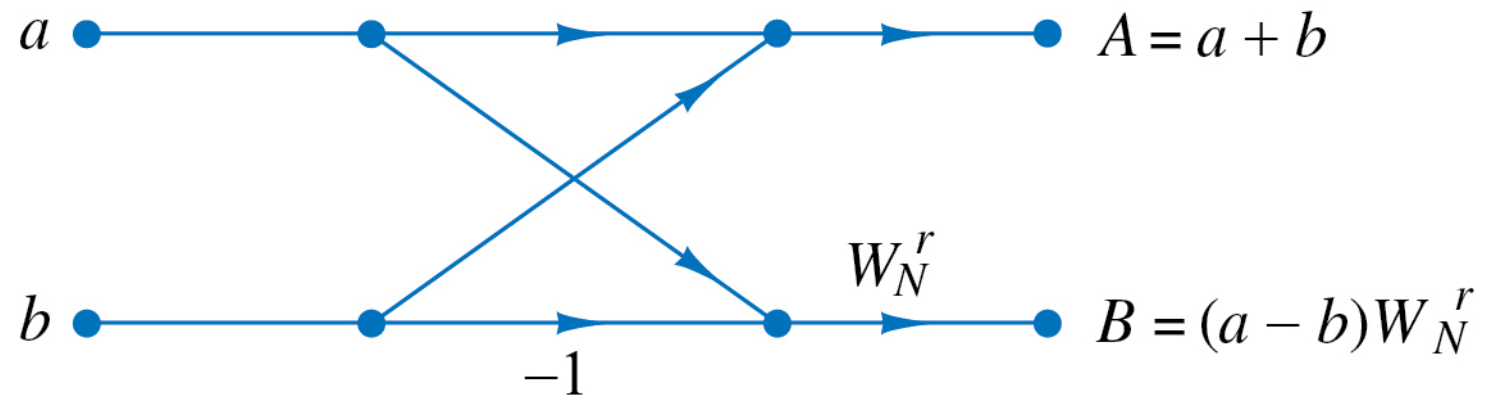


Figure 8.1.11 $N = 8$ -point decimation-in-frequency FFT algorithm.

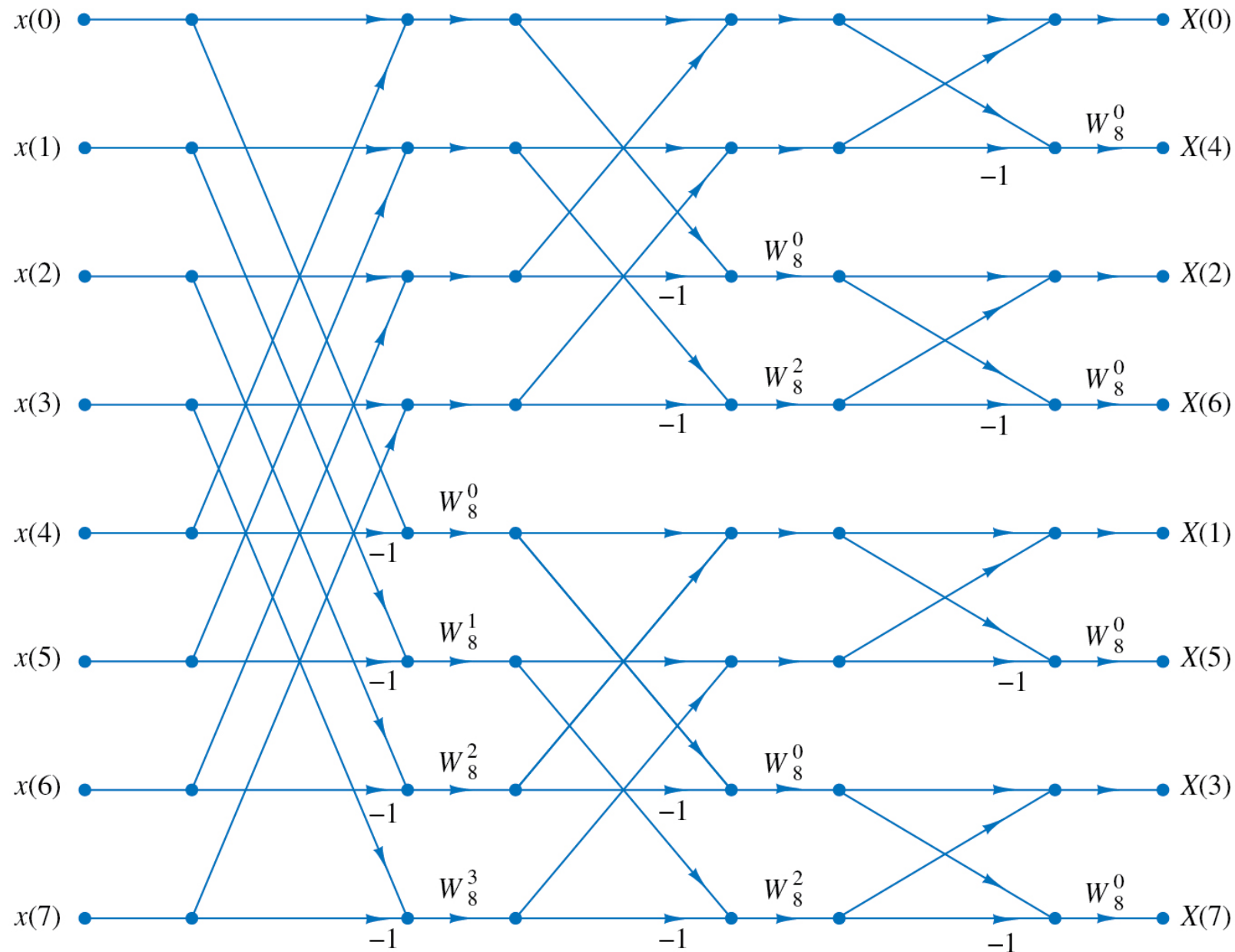


Figure 8.1.12 Basic butterfly computation in a radix-4 FFT algorithm.

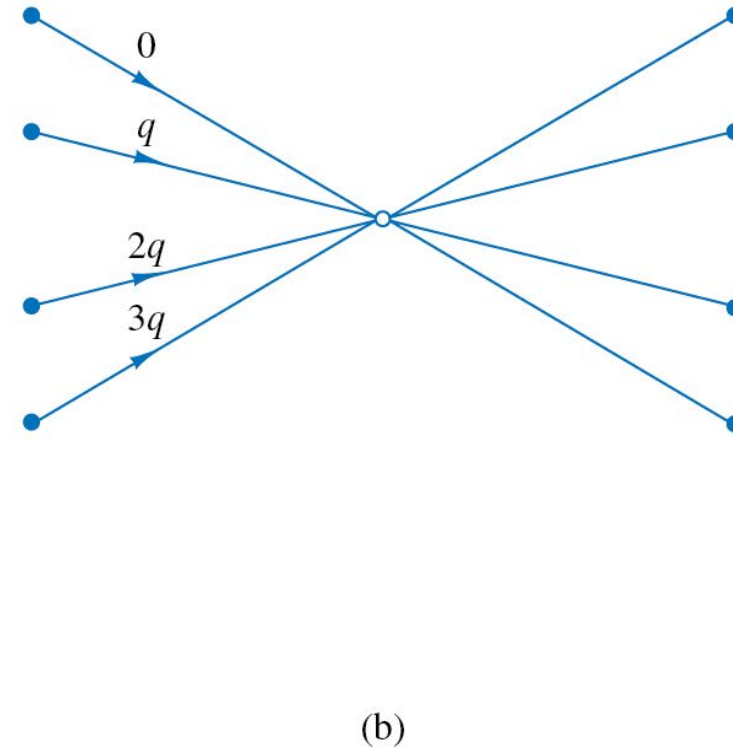
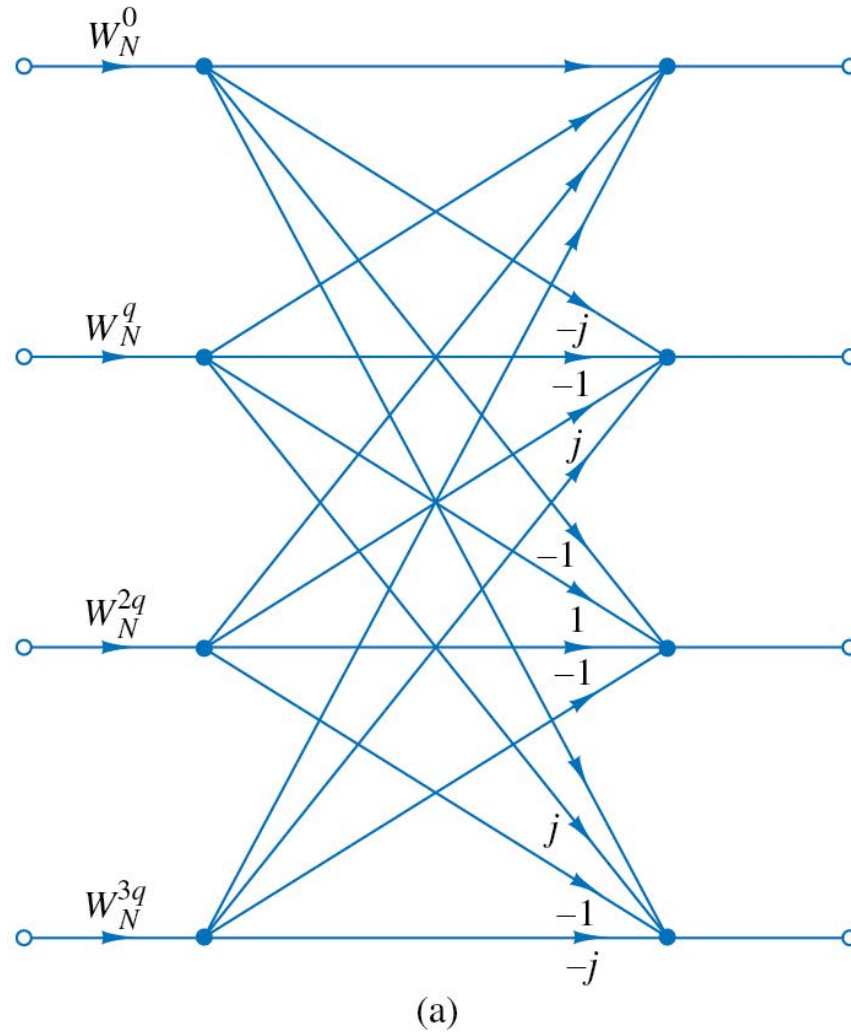


Figure 8.1.13 Sixteen-point radix-4 decimation-in-time algorithm with input in normal order and output in digit-reversed order. The integer multipliers shown on the graph represent the exponents on W_{16} .

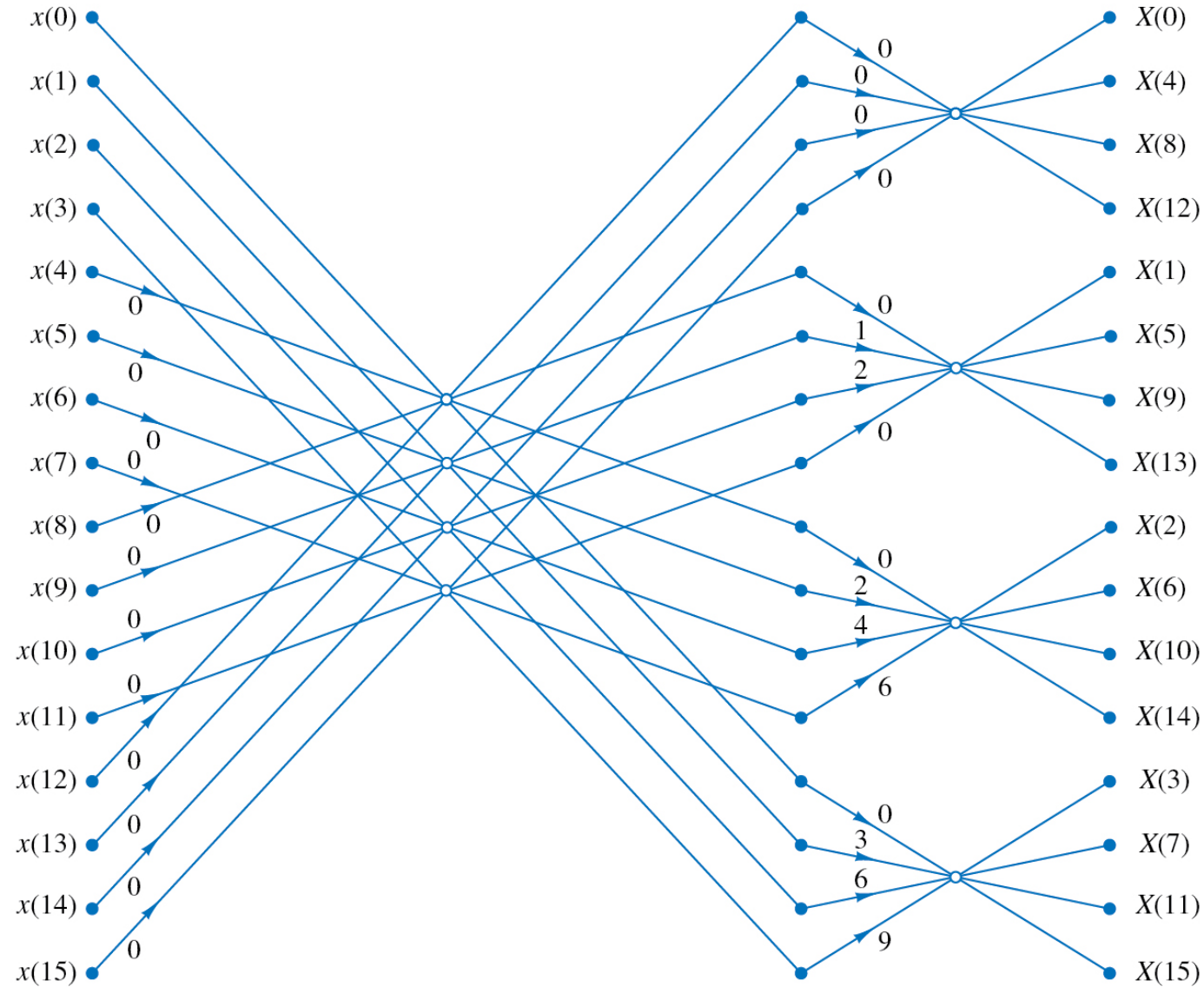


Figure 8.1.14 Sixteen-point, radix-4 decimation-in-frequency algorithm with input in normal order and output in digit-reversed order.

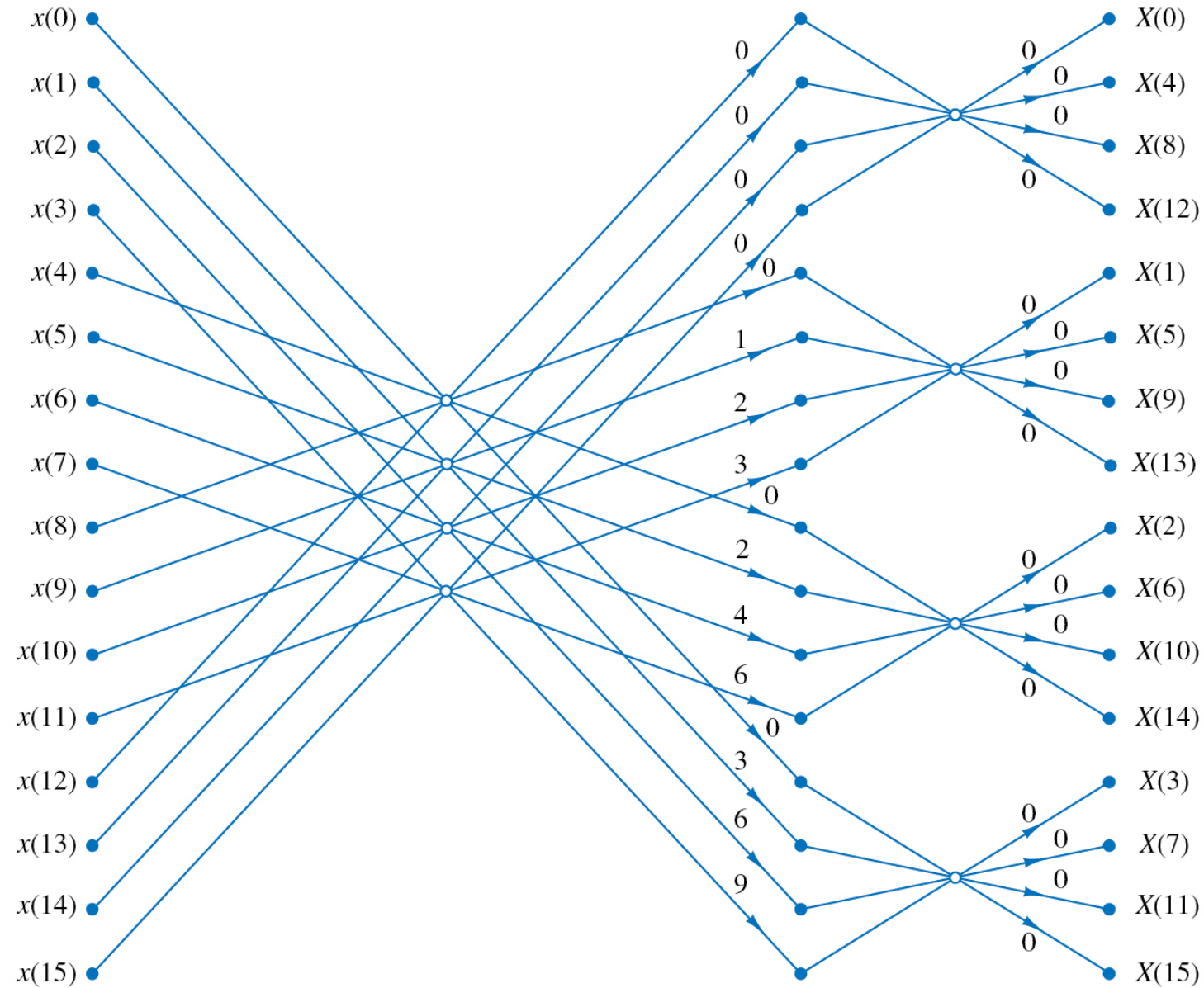


Figure 8.1.15 Length 32 split-radix FFT algorithms from paper by Duhamel (1986); reprinted with permission from the IEEE.

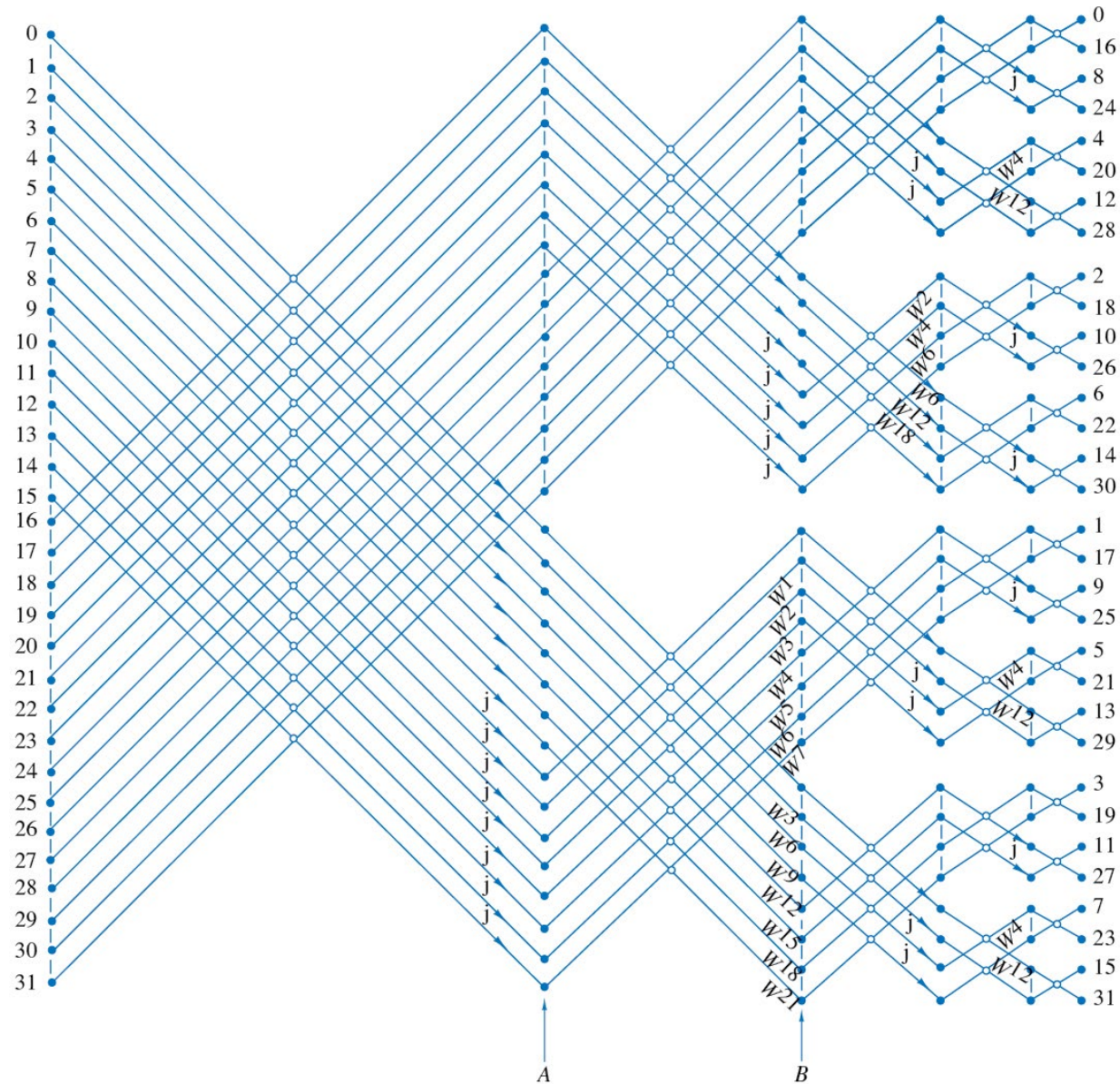


Figure 8.1.16 Butterfly for SRFFT algorithm.

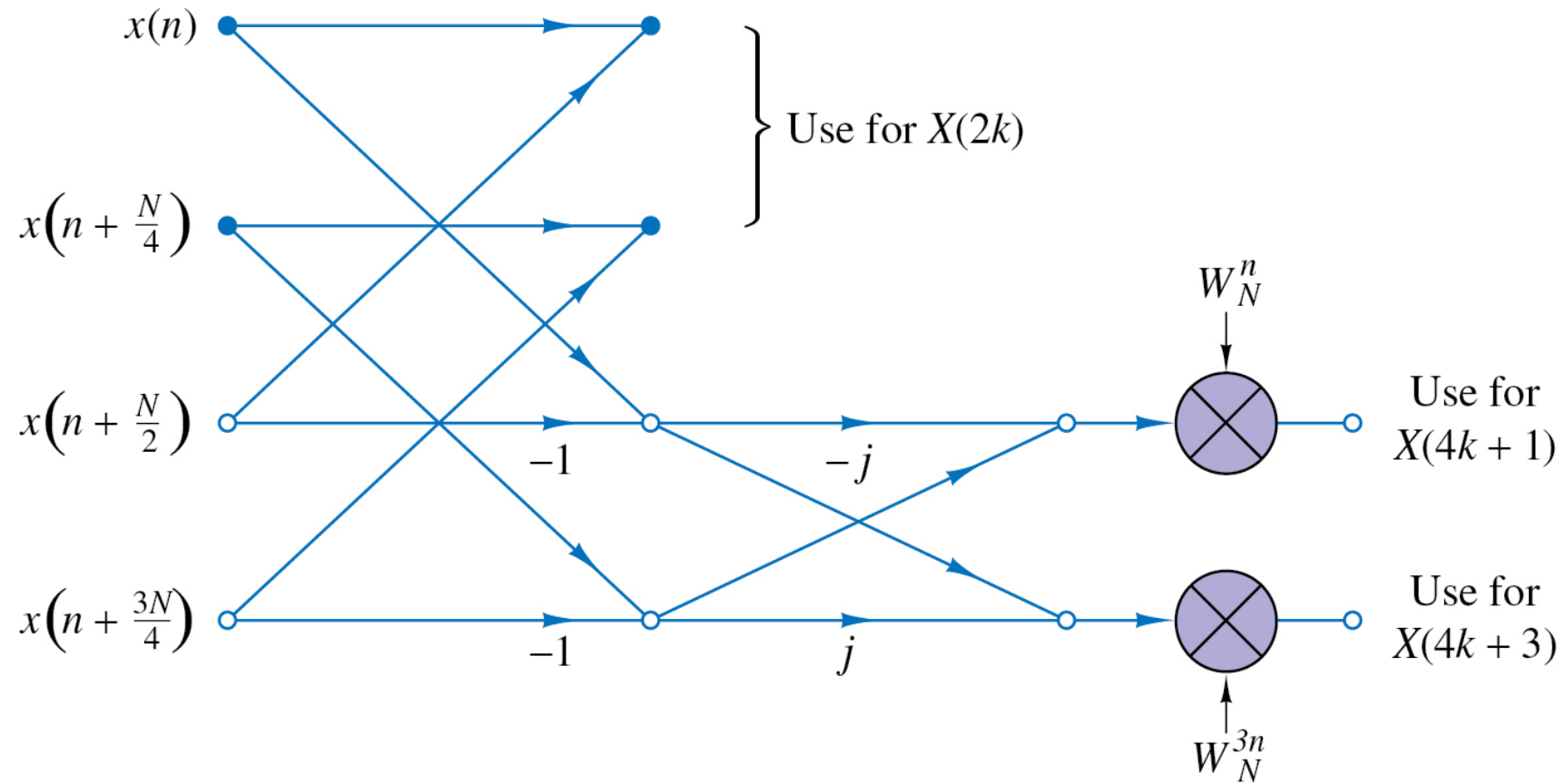


Table 8.2 Number of Nontrivial Real Multiplications and Additions to Compute an N -point Complex DFT

N	Real Multiplications				Real Additions			
	Radix 2	Radix 4	Radix 8	Split Radix	Radix 2	Radix 4	Radix 8	Split Radix
16	24	20		20	152	148		148
32	88			68	408			388
64	264	208	204	196	1,032	976	972	964
128	712			516	2,504			2,308
256	1,800	1,392		1,284	5,896	5,488		5,380
512	4,360		3,204	3,076	13,566		12,420	12,292
1,024	10,248	7,856		7,172	30,728	28,336		27,652

Source: Extracted from Duhamel (1986).

Table 8.3 Computational Complexity

Size of FFT $v = \log_2 N$	$c(v)$ Number of Complex Multiplications per Output Point
9	13.3
10	12.6
11	12.8
12	13.4
14	15.1

Summary

References

- [DSP] Proakis, J.G. and Manolakis, D.G., Digital Signal Processing: Principles, Algorithms, and Applications, Pearson, 5th ed., 2022.

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